THE ASSAM
ROYAL GLOBAL UNIVERSITY
GUWAHATI

Royal School of Engineering and Technology (RSET)

Department of Computer Science and Engineering (CSE)

Course Structure & Syllabus

(Based on National Education Policy 2020)

For

Bachelor of Technology

In

Computer Science and Engineering

W.E.F

AY: 2026-2027

Section 6

Course Structure and Syllabus of the Framework

6.1 Course Structure of B. Tech. (CSE)

1st Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Basic Science Course (BSC)							
1	PHY022C101	Physics	3	0	0	3	3
2	PHY022C111	Physics Lab	0	0	2	1	2
3	MAT022C102	Mathematics – I	3	1	0	4	4
Engineering Science Course (ESC)							
4	CSE022C103	Basic Electrical Engineering	3	0	0	3	3
5	CSE022C113	Basic Electrical Engineering Lab	0	0	2	1	2
6	CEE022C114	Engineering Graphics & Design Lab	0	0	4	2	4
7	CSE022C105	Programming for Problem Solving -I	2	0	0	2	2
8	CSE022C115	Programming for Problem Solving -I Lab	0	0	2	1	2
Humanities/Social Science including Management Course (HSMC)							
9	BHS022A101	Universal Human Values: Understanding Harmony and Ethical Human Conduct	3	0	0	3	3
Mandatory Courses (MC)/ Skill Enhancement Courses							
10	CSE022S116	Design Thinking and Ideation Lab	0	0	2	1	2
		TOTAL	**14**	**1**	**12**	**21**	**27**
2nd Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Basic Science Course (BSC)							
1	CHY022C201	Chemistry	3	0	0	3	3
2	CHY022C211	Chemistry Lab	0	0	2	1	2
3	MAT022C202	Mathematics – II	3	1	0	4	4
4	CSE022C203	Basic Environmental Engineering	3	0	0	3	3
Engineering Science Course (ESC)							
5	CSE022C204	Programming for Problem Solving-II	3	0	0	3	3
6	CSE022C214	Programming for Problem Solving Lab	0	0	2	1	2
7	MEE022C215	Manufacturing Practices Workshop	0	0	4	2	4
Humanities/Social Science including Management Course (HSMC)							
8	CEN022A201	English for Technical Writing	2	0	0	2	2
Mandatory Courses (MC)/ Skill Enhancement Courses							
9	CSE022S217	Sports and Yoga	0	0	2	1	2
		TOTAL	**14**	**1**	**10**	**20**	**25**
3rd Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**

Professional Core Course (PCC)							
1	MAT022C301	Discrete Mathematics	3	1	0	4	4
2	CSE022C302	Data Structures and Algorithms	3	0	0	3	3
3	CSE022C312	Data Structures and Algorithms Lab	0	0	2	1	2
4	CSE022C303	Computer Organisation and Architecture	3	0	0	3	3
Engineering Science Course (ESC)							
5	CSE022C304	Digital Logic and Design	3	0	0	3	3
6	CSE022C314	Digital Logic and Design Lab	0	0	2	1	2
IKS							
7	IKS992K301	Indian Knowledge System-I	2	0	0	2	2
Open Elective							
8	CSE022C305	OOP using C++	3	0	0	3	3
9	CSE022C315	OOP using C++ Lab	0	0	2	1	2
Internship							
10	CSE022C327	Internship-I	0	0	0	2	0
		TOTAL	**17**	**1**	**6**	**23**	**24**
4th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C401	JAVA Programming	3	0	0	3	3
2	CSE022C411	JAVA Programming Lab	0	0	2	1	1
3	CSE022C402	Database Management Systems	3	0	0	3	3
4	CSE022C412	Database Management Systems Lab	0	0	2	1	2
5	CSE022C403	Formal Language and Automata Theory	3	1	0	4	4
Engineering Science Course (ESC)							
6	CSE022C404	Microprocessor	3	0	0	3	3
7	CSE022C414	Microprocessor Lab	0	0	2	1	2
Open Elective							
8	CSE022C405	Programming with Python	3	0	0	3	3
9	CSE022C415	Programming with Python Lab	0	0	2	1	1
Internship							
10	CSE022C427	Internship-II	0	0	0	2	0
		TOTAL	**15**	**1**	**8**	**22**	**24**
5th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C501	Operating Systems	3	0	0	3	3
2	CSE022C511	Operating Systems Lab	0	0	2	1	2
3	CSE022C502	Introduction to Data Science	3	0	0	3	3
4	CSE022C512	Introduction to Data Science Lab	0	0	2	1	2
5	CSE022C503	Design and Analysis of Algorithms	3	0	0	3	3
6	CSE022C513	Design and Analysis of Algorithms Lab	0	0	2	1	2

Engineering Science Course (ESC)							
6	CSE022C504	Data Communication	3	0	0	3	3
Humanities/Social Science including Management Course (HSMC)							
7	BSA022C505	Principles of Management & Organisational Behaviour	3	0	0	3	3
Open Elective							
8	CSE022G506	Open Elective-III (Prompt Engineering)	3	0	0	3	3
Internship							
9	CSE022C527	Internship-III	0	0	0	2	0
		TOTAL	**18**	**0**	**6**	**23**	**24**
6th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C601	Computer Networks	3	0	0	3	3
2	CSE022C611	Computer Networks Lab	0	0	2	1	2
3	CSE022C602	Compiler Design	3	0	0	3	3
4	CSE022C612	Compiler Design Lab	0	0	2	1	2
5	CSE022C603	Software Engineering	3	0	0	3	3
Professional Elective Course (PEC)							
6	CSE022D60X	Professional Elective Course-I	4	0	0	4	4
7	CSE022D60X	Professional Elective Course-II	4	0	0	4	4
Open Elective							
8	CSE022G606	Open Elective-IV (Entrepreneurship and Startup)	3	0	0	3	3
Internship							
9	CSE022C627	Internship-IV	0	0	0	2	0
		TOTAL	**20**	**0**	**4**	**24**	**24**
7th Semester							
S. No.	**Subject Code**	**Names of subjects**	**L**	**T**	**P**	**C**	**TCP**
Professional Core Course (PCC)							
1	CSE022C701	Introduction to Artificial Intelligence	3	0	0	3	3
2	CSE022C711	Introduction to Artificial Intelligence Lab	0	0	2	1	2
3	CSE022C702	Web Technology	3	0	0	3	3
4	CSE022C712	Web Technology Lab	0	0	2	1	2
Professional Elective Course (PEC)							
5	CSE022D70X	Professional Elective Course-III	4	0	0	4	4
6	CSE022D70X	Professional Elective Course-IV	4	0	0	4	4
Internship							
7	CSE022C725	Internship-V	0	0	0	2	0
Project/ Dissertation							
8	CSE022C726	Project-I	0	0	6	2	8
		TOTAL	**13**	**0**	**2**	**20**	**25**
8th Semester							

S. No.	Subject Code	Names of subjects	L	T	P	C	TCP
Professional Core Course (PCC)							
1	CSE022C801	Introduction to Machine Learning	3	0	0	3	3
2	CSE022C811	Introduction to Machine Learning Lab	0	0	2	1	2
3	CSE022C802	Cryptography and Network Security	3	0	0	3	3
4	CSE022C812	Cryptography and Network Security Lab	0	0	2	1	2
Professional Elective Course (PEC)							
5	CSE022D80X	PEC-V	4	0	0	4	4
Project/ Dissertation							
6	CSE022C824	Project-II	0	0	12	6	8
		TOTAL	**11**	**0**	**14**	**18**	**21**

PEC Tracks	Course Codes	Subjects
Network Engineering	CSE022D601	PEC 1: Mobile Computing
	CSE022D602	PEC 2: Big Data Analytics
	CSE022D701	PEC 3: Cloud Computing
	CSE022D702	PEC 4: Wireless Sensor Networks
	CSE022D801	PEC 5: Internet of Things
Semiconductors	CSE022D603	PEC 1: Introduction to Semiconductor Devices
	CSE022D604	PEC 2: VLSI Design
	CSE022D703	PEC 3: Embedded Systems and IOT
	CSE022D704	PEC 4: Micro Electro Mechanical Systems / VHDL
	CSE022D802	PEC 5: Nanoelectronics
General Track	CSE022D605	PEC 1: Social Network Analysis
	CSE022D606	PEC 2: Digital Image Processing
	CSE022D705	PEC 3: Introduction to Data Science
	CSE022D706	PEC 4: Fundamentals of Quantum Computing
	CSE022D801	PEC 5: Introduction to Cyber Security

List of Open Electives to be offered by Department of CSE	
Open Elective-I	Programming with Python (CSE022G306)
Open Elective-II	Fundamentals of Web Design (CSE022G406)
Open Elective-III	Introduction to AI (CSE022G506)/ Prompt Engineering
Open Elective-IV	Fundamentals of IOT (CSE022G606)

6.2 Detailed Syllabus of 1st Semester

Paper I/Subject Name: Physics		Subject Code: PHY022C101
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make the students enhance the fundamental knowledge in Physics and its applications relevant to various streams of Engineering and Technology

Prerequisites: Concepts of Physics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic terminologies of physics	**BT 1**
CO 2	**Understand** the basic concepts of Physics	**BT 2**
CO 3	**Solve** problems in wave mechanics.	**BT 3**
CO 4	**Analyze** knowledge in calculating red and blue shift and also in acoustics.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Electromagnetic Theory**	Electrostatics in vacuum, Electrostatics in a linear dielectric medium, Magneto statics, Magneto statics in a linear magnetic medium, Faraday's law, Displacement current, Magnetic field due to time-dependent electric field and Maxwell's equations, Electromagnetic waves	**13**
II.	**Introduction to Mechanics**	Transformation of scalars and vectors under Rotation transformation; Potential energy function; Non-inertial frames of reference; Harmonic oscillator; Definition and motion of a rigid body in the plane; Rotation in the plane; Kinematics in a coordinate system rotating and translating in the plane; Angular momentum, Introduction to three-dimensional rigid body motion	**25**
III.	**Quantum Mechanics for Engineers**	Wave nature of particles and the Schrodinger equation, Mathematical Preliminaries for quantum mechanics, Applying the Schrodinger equation, molecular bonding, Solids	**10**
IV	**Oscillations, Waves and Optics**	Simple harmonic motion, damped and forced simple harmonic oscillator, Non-dispersive transverse and longitudinal waves in one dimension and introduction to dispersion, The propagation of light and geometric optics, Wave optics, Lasers	**18**
		TOTAL	**66**

Physics Lab Syllabus

Detailed Syllabus:

Experiment	Experiment Title	Lab Hours
I	Determination of Moment of Inertia of a given solid about its own axis by using M.I.Table	2
II	Determination of Young's Modulus using Searle's Apparatus	2
III	Determination of Rigidity of Modulus of the material of the given rod by Statiscal method	2
IV	Determination of Powers of Given lenses using an Optical Bench i. Concave Lens, ii Convex Lens	2
V	Determination of Resistance of a Galvanometer using Post Office Box.	2
VI	To determine the mechanical equivalent of heat by Joule's calorimeter	2
VII	Determination of ratio of E.M.F of two cells using Potentiometer.	2
VIII	To determination of the focal length of a convex mirror with the help of an auxiliary lens.	2
IX	Determination of Horizontal Components of Earth's Magnetic field using Magnetometer	2
X	Determination of coefficient of Viscosity of water by Capillary Flow Method	2

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Elements of properties of matter,* Mathur D.S., 7th Edition, Revised Edition, 2005, S. Chand publication, New Delhi.
2. *Electricity and Magnetism,* Tayal D.C, Publisher, 4th Edition, 2017, Himalaya Publishing House,New Delhi.
3. *Geometrical and Physical Optics,* Chakraborty P.K., 3rd Edition, 2005, New Central Book agency (P) Ltd.

Reference Books:

1. Singh A.K. and Malik Hitendra *Engineering Physics,* 2nd Edition, 2016, McGraw Hill Education Private Limited. New Delhi.
2. Gaur R.K and Gupta S.L, *Engineering Physics*, 2015, Dhanpat Rai publication, New Delhi.
3. Arthur Beiser, Shobhit Mahajan, S. Rai. Choudhury, *Concept of Modern physics,* 6th Edition, 2009, McGraw-Hill education Private limited. New Delhi.
4. M Ghosh & D Bhattacharya, *A Textbook of Oscillations, Waves and Acoustics,* 5th Edition, 2016, S. Chand publication.

Additional Readings

1. https://www.griet.ac.in/nodes/Engineering%20Physics%20Notes.pdf
2. https://mrcet.com/downloads/digital_notes/HS/R20/Engineering%20Physics.pdf
3. NPTEL Course on Introduction To Electromagnetic Theory by Prof. Manoj Harbola, IIT Kanpur

Paper II/Subject Name: Mathematics-I		**Subject Code: MAT022C102**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to enable students to achieve conceptual understanding and to retain the best traditions of traditional calculus

Prerequisites: Concepts of Mathematics of +2 level

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic terminologies of calculus	**BT 1**
CO 2	**Understand** the applications of differential and integral calculus in different fields of Engineering.	**BT 2**
CO 3	**Apply** the single and multivariable differential and Integral calculus in engineering problems.	**BT 3**
CO 4	**Analyze** and **assess** the patterns in series	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I.	**Basic & Single Variable Calculus**	Curvature, evolutes and involutes; Evaluation of definite and improper integrals; Beta and Gamma functions and their properties; Applications of definite integrals to evaluate surface areas and volumes of revolutions. Rolle's Theorem, Mean value theorems and applications; Extreme values of functions; Linear approximation; Indeterminate forms and L' Hospital's rule.	**15**
II.	**Multi Variable Calculus I**	Limit, continuity and partial derivatives, directional derivatives, gradient, total derivative; Tangent plane and normal line; Maxima, minima and saddle points; Method of Lagrange multipliers.	**15**
III.	**Multi Variable Calculus II**	Multiple Integration: Double integrals (Cartesian), change of order of integration in double integrals, Change of variables (Cartesian to polar), Applications: areas and volumes, Center of mass and Gravity (constant and variable densities); Triple integrals (Cartesian), orthogonal curvilinear coordinates, Simple applications involving cubes, sphere and rectangular parallelepipeds; Scalar line integrals, vector line integrals, scalar	**15**

		surface integrals, vector surface integrals, Gradient, curl and divergence, Theorems of Green, Gauss and Stokes.	
IV	**Sequence and Series**	Limits of sequence of numbers, Calculation of limits, Infinite series; Tests for convergence; Power series, Taylor and Maclaurin series; Taylor theorem, convergence of Taylor series, error estimates.	**15**
		TOTAL	**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4*15 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *A text book of Engineering Mathematics*, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. *Mathematical Methods for Physics and Engineering: A Comprehensive Guide,* K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics*, 43rd Edition, 2014, Khanna Publishers.
2. Das B. C. & Mukherjee B. N., *Differential Calculus*, 55th Edition, U. N. Dhur & Sons Pvt. Ltd.
3. Das B. C. & Mukherjee B. N, *Integral Calculus*, 57th Edition, U. N. Dhur & Sons Pvt. Ltd

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Mathematics-I.
2. https://www.vidyalankar.org/gate/assets/docs/notes/maths.pdf

Paper III/Subject Basic Electrical Engineering		**Subject Code: CSE022C103**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make students understand the basic electrical terminologies and familiarize them with the basic concepts of Electronic Circuits,D.C. networks, single-phase & three-phase A.C. networks and Electrical Machines

Prerequisites: Basic concepts of D.C. networks of Class XII, Electromagnetic Induction and A.C. Fundamentals etc.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Recall** the basic concepts of electrical engineering	**BT 1**
CO 2	**Understand** the concept behind basic electric and electroniccircuits.	**BT 2**
CO 3	**Analyze** the working principles of Electrical Machines.	**BT 3**
CO 4	**Apply** the principles of electronic devices to analyze and design basic rectifier and electronic circuits.	**BT 4**

Detailed Syllabus:

Modules	Topic	Course Content	Hours
I.	**DC Circuits**	Electrical Circuit Elements – The resistance element, the inductance element, the capacitance element. Voltage & Current sources, practical & ideal voltage and current sources, source transformation. Kirchhoff's Laws, Analysis of simple circuits with DC excitation – series circuit, parallel circuit, voltage and current divider rule, star -delta conversion, Maxwells mesh current method, Nodal voltage analysis. Network Theorems- Thevenin's Theorem, Superposition Theorem	**12**
II.	**AC Circuits**	AC fundamentals – generation of alternating voltage, representation of sinusoidal waveform, concept of frequency, cycle, time period, instantaneous value, average value, peak value, RMS value, phasor representation. Single phase AC Circuits – analysis of single-phase AC circuits consisting of R-L-C parameters, apparent power, real power, reactive power, power factor and its significance. Analysis of R-C, R-L & R-L-C series circuit. Introduction to 3-phase AC circuits-Star & Delta configuration.	**20**
III.	**Introduction to Electrical Machines:**	Principle of operation and construction of single-phase transformers. EMF equation, losses in transformers DC Machines – Constructional details of a DC Machine; EMF Equation of a DC machine, types of DC Machines. Operation of a DC machine as a motor, Importance of back emf. Principle of operation and construction of single-phase transformers. EMF equation, losses, efficiency and voltage regulation.	**20**

Modules	Topic	Course Content	Hours
IV.	**Basic Electronics**	Semiconductors: p-type & n-type, pn junction, Bias-Forward & Reverse Bias, V-A Characteristics of pn junction, Semiconductor Diode, V-I characteristics of Diode, Introduction to rectification, half-wave, full-wave and bridge rectifiers. Earthing – difference between neutral wire & earth wire, methods of earthing of domestic fittings and appliances. Electrical Power Supply System; Three phase systems: Delta & Star configuration; Phasor diagrams and numericals on the 3-phase systems Protection of electrical installations against overload, short circuit and earth fault. Protective devices for overload, short circuit, earth fault and electric shock – SFU, MCB, ELCB. Types of wires, cables and wiring used in electrical installations	**14**
		TOTAL	**66**

Basic Electrical Engineering Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

Lab	Experiments	Hours
I	To verify Thevenin's Theorem for DC network	**2**
II	To verify Maximum Power Transfer Theorem for DC network	**2**
III	Study of R-L-C Series circuit and determine R,L,C, cos Φ,P and Q and draw the phasor diagram	**2**
IV	Study of R-L-C Parallel circuit and determine R,L,C,cos Φ,P and Q and draw the phasor diagram	**2**
V	Calibration of a milli-ammeter as a voltmeter.	**2**
VI	To determine the ohmic and effective resistance (armature winding)	**2**
VII	To study the characteristics of a filament lamp	**2**
VIII	To measure the power in a single-phase load using one wattmeter	**2**
IX	To measure the insulation resistance using Megger	**2**
X	Demonstration of house wiring	**2**
	TOTAL	**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**

3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A Text Book of Electrical Technology*, Thereja, B.L., 1st Edition revised, 2008, S Chand & Company Ltd. Ram Nagar; New Delhi.
2. *Basic Electrical Engineering*, D. P. Kothari, I. J.Nagrath, 3rd Edition, 2009, Tata McGraw-Hill

Reference Books:

1. D. C. Kulshreshtha, *Basic Electrical Engineering*, 1st Edition, 2009, McGraw-Hill
2. E. Hughes, *Electrical and Electronics Technology,* 10th Edition, 2011, Pearson Publication

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Basic%20Electrical%20Engineering%20R-20.pdf
2. https://www.cet.edu.in/noticefiles/231_BASIC_ELECTRICAL_ENGG-min.pdf
3. NPTEL Course on Basic Electrical Circuits by Prof. Nagendra Krishnapura, IITM
4. NPTEL Course on Fundamentals of Electrical Engineering by Prof. Debapriya Das, IIT, Kharagpur

Paper IV/Subject Name: Engineering Graphics & Design Lab		**Subject Code: CEE022C114**
L-T-P-C – 0-0-4-2	**Credit Units: 02**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make students understand the process of drawing projections and sections and basic engineering drawing formats and to convert sketches to engineered drawings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Relate** with the concepts of drawings and projections	**BT 1**
CO 2	**Understand** the dimension and figures using the drawing instruments and acquire visualization skills, projection of points, etc.	**BT 2**
CO 3	**Utilize** engineering curves in tracing the paths of simple machine components.	**BT 3**
CO 4	**Analyze** and assess sketches to convert them to engineered drawings.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I.	**Introduction and Projections**	Principles of Engineering Graphics and their significance, usage of Drawing instruments, lettering, Conic sections including the Rectangular Hyperbola (General method only); Cycloid, Epicycloid, Hypocycloid and Involute; Scales – Plain, Diagonal and Vernier Scales; Principles of Orthographic Projections-Conventions - Projections of Points and lines inclined to both planes; Projections of planes inclined Planes - Auxiliary Planes; Covering those inclined to both the Planes- Auxiliary Views; Draw simple annotation, dimensioning and scale. Floor plans that include: windows, doors, and fixtures such as WC, bath, sink, shower, etc	**11**
II.	**Angular Solids and Isometric Projections**	Prism, Cylinder, Pyramid, Cone – Auxiliary Views; Development of surfaces of Right Regular Solids - Prism, Pyramid, Cylinder and Cone; Draw the sectional orthographic views of geometrical solids, objects from industry and dwellings (foundation to slab only). Principles of Isometric projection – Isometric Scale, Isometric Views, Conventions; Isometric Views of lines, Planes, Simple and compound Solids; Conversion of Isometric Views to Orthographic Views and Vice-versa, Conventions;	**11**
III.	**Overview of Computer Graphics**	Listing the computer technologies that impact on graphical communication, Demonstrating knowledge of the theory of CAD software [such as: The Menu System, Toolbars (Standard, Object Properties, Draw, Modify and Dimension), Drawing Area (Background, Crosshairs, Coordinate System), Dialog boxes and windows, Shortcut menus (Button Bars), The Command Line (where applicable), The Status Bar, Different methods of zoom as used in CAD, Select and erase objects.; Isometric Views of lines, Planes, Simple and compound Solids]. Consisting of set up of the drawing page and the printer, including scale settings, setting up of Modules and drawing limits; ISO and ANSI standards for coordinate dimensioning and tolerancing; Orthographic constraints, Snap to objects manually and automatically; Producing drawings by using various coordinate input entry methods to draw straight lines, Applying various ways of drawing circles;	**11**
IV	**Customisation and CAD drawing**	Covering applying dimensions to objects, applying annotations to drawings; Setting up and use of Layers, layers to create drawings, Create, edit and use customized layers; Changing line lengths through modifying existing lines (extend/lengthen); Printing documents to paper using the print command; orthographic projection techniques; Drawing sectional views of composite right regular geometric solids and project the true shape of the sectioned surface; Drawing annotation, Computeraided design (CAD) software modeling of parts and assemblies. Parametric and non-parametric solid, surface, and wireframe models. Part editing and two-dimensional documentation of models. Planar projection theory, including sketching of perspective, isometric, multiview, auxiliary, and section views. Spatial visualization exercises. Dimensioning guidelines, tolerancing techniques; dimensioning and scale multi views of dwelling;	**11**
		TOTAL	**44**

Credit Distribution

Lecture/ Tutorial	Practicum	Experiential Learning
--	2 * 15 NCH = 30 NCH	30 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Engineering Drawing;* Bhatt, N.D, 53rd Edition, 2016, Charotar Publishing House

Reference Books:

1. Jolhe Dhananjay A; *Engineering drawing*, 5th Edition, 2010, Tata McGraw-Hill Education Pvt. Ltd., New Delhi

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Engineering%20Graphics%20Manual%20final.pdf
2. https://www.pvpsiddhartha.ac.in/autonomus14/1-1/it/IT1L3.pdf
3. NPTEL Course on Engineering Drawing and Computer Graphics by Prof. Rajaram Lakkaraju, IIT, Kharagpur
4. NPTEL Course on Engineering Graphics by Prof. Nihar Ranjan Patra, IIT, Kanpur

Paper V/Subject Name: Programming for Problem Solving-I		**Subject Code: CSE022C105**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students develop computational thinking and problem-solving skills, learn the fundamentals of C programming, design algorithms and implement solutions using control structures, arrays and strings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the various constructs used in programming	**BT 1**
CO 2	**Explain** the fundamental concepts of programming, algorithms, flowcharts, data types, operators, and control structures in C.	**BT 2**
CO 3	**Develop** C programs using decision-making and iterative constructs to solve computational problems	**BT 3**
CO 4	**Analyze** and debug the errors while writing the programs.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Fundamentals of Programming**	Introduction to components of a computer system (disks, memory, processor, where a program is stored and executed, operating system, compilers etc.) Idea of Algorithm: steps to solve logical and numerical problems. Representation of Algorithm: Flowchart/Pseudocode with examples. From algorithms to programs; source code, variables (with data types) variables and memory locations, Syntax and Logical Errors in compilation, object and executable code.	**15**
II	**C Fundamentals**	Structure of a C program, Character set, tokens, keywords and identifiers, Variables and constants, Data types, Operators and expressions, Input and output functions, Type conversion, Compilation and execution of C programs	**15**
III	**Decision Making and Iterative Programming**	Conditional statements: if, if-else, nested if, else-if ladder, switch. Looping constructs: while, do-while, for, Nested loops, break, continue and goto. Programming using loops, Pattern generation, Number-based applications, Programs based on condition and looping statements	**15**
IV	**Arrays, Strings and Basics of Functions**	One-dimensional arrays: declaration, initialisation, Two-dimensional arrays, Matrix operations, Character arrays, Strings, Standard string library functions, Applications of arrays and strings, Searching in arrays, Sorting techniques, Programming exercises Introduction to User defined functions: function prototype, function call, function declaration, call by value, call by reference	**15**
TOTAL			**60**

Programming for Problem Solving-I Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Character set, Tokens, Keywords and Identifiers, Constants, variables, data types, statements, comments, declaration of storage class, assigning values to variables.
2. Managing I/O, reading and writing characters, formatted Input/output.
3. Arithmetic operators, relational operators, logical operators, assignment operators, increment & decrement operators, conditional operators, bitwise operators, special operators.
4. Importance of decision making, decision making with if statement, if-else statement, nested if-else statements, switch-case statement.
5. Importance of iterative statements, the while statement, do-while statement, for statement, nested for looping.
6. Significance of Arrays, creation and use of one & two-dimensional arrays
7. Declaration and use of string variables, reading and writing strings.
8. User defined functions

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
2. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur

Paper V/Subject Name: Universal Human Values		**Subject Code: BHS022A101**
L-T-P-C – 2-0-0-2	**Credit Units: 02**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to help the students appreciate the essential complementarily between 'VALUES' and 'SKILLS' to ensure sustained happiness and prosperity which are the core aspirations of all human beings.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic need of human values in real life	**BT 1**
CO 2	**Understand** the importance of following the basic universal human values	**BT 2**
CO 3	**Apply** the holistic understanding in one's day-today life so as to keep oneself happy and to socialize with nature, society, etc	**BT 3**
CO 4	**Analyze** the harmony within human beings by distinguishing the needs of the self and the body.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Value Education**	Right Understanding, Relationship and Physical Facility	**11**

		(Holistic Development and the Role of Education), Understanding Value Education, Sharing about Oneself, Self-exploration as the Process for Value Education, Continuous Happiness and Prosperity – the Basic Human Aspirations, Exploring Human Consciousness, Happiness and Prosperity – Current Scenario, Lectured, Method to fulfil the Basic Human Aspirations, Exploring Natural Acceptance	
II	**Harmony in Human Being**	Understanding Human being as the Co-existence of the Self and the Body, distinguishing between the Needs of the Self and the Body, Exploring the difference of Needs of Self and Body, The Body as an Instrument of the Self Understanding Harmony in the Self, Exploring Sources of Imagination in the Self, Harmony of the Self with the Body, Programme to ensure self-regulation and Health, Exploring Harmony of Self with the Body	**11**
III	**Harmony in the Family & Society**	Harmony in the Family – the Basic Unit of Human Interaction, 'Trust' – the Foundational Value in Relationship", Exploring the Feeling of Trust, 'Respect' – as the Right Evaluation, Exploring the Feeling of Respect, Other Feelings, Justice in Human-to-Human Relationship Understanding Harmony in the Society, Vision for the Universal Human Order, Exploring Systems to fulfil Human Goal	**11**
IV	**Harmony in Nature & Implications of Holistic Understanding**	Understanding Harmony in the Nature, Interconnectedness, self-regulation and Mutual Fulfilment among the Four Orders of Nature, Exploring the Four Orders of Nature, Realizing Existence as Co-existence at All Levels, The Holistic Perception of Harmony in Existence, Exploring Co-existence in Existence Natural Acceptance of Human Values, Definitiveness of (Ethical) Human Conduct, Exploring Ethical Human Conduct, A Basis for Humanistic Education, Humanistic Constitution and Universal Human Order, Competence in Professional Ethics, Exploring Humanistic Models in Education, Holistic Technologies, Production Systems and Management Models-Typical Case Studies, Strategies for Transition towards Value-based Life and Profession Exploring Steps of Transition towards Universal Human Order	**11**
TOTAL			**44**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2 * 22 NCH = 44 NCH	-	8 * 2 NCH = 16 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *A Foundation Course in Human Values and Professional Ethics,* R R Gaur, R Asthana, G P Bagaria, 2nd Revised Edition, Excel Books, New Delhi

Reference Books:

1. *Human Values*, A.N. Tripathi, 3rd Edition, 2019, New Age Intl. Publishers, New Delhi,

Additional Readings:

1. https://uhv.org.in/uhv2notes
2. https://fdp-si.aicte-india.org/UHV-II%20Class%20Note.php

Paper VI/Subject Name: Design Thinking and Ideation Lab		**Subject Code: CSE022S116**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students understand the principles and process of design thinking, develop empathy for users and identify real-world problems, apply creative thinking and ideation techniques to generate innovative solutions, build and test prototypes through iterative design, and work collaboratively to develop user-centric solutions.

Prerequisites: None

Course Outcomes:

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Tell** the utility of design thinking and ideation lab	**BT 1**
CO 2	**Explain** the stages and principles of the design thinking process for solving user-centric problems	**BT 2**
CO 3	**Apply** empathy, observation, and problem-definition techniques to identify real-world challenges.	**BT 3**
CO 4	**Analyze** user requirements and generate innovative solutions using structured ideation techniques.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 22 (2 hours per week)

Minimum 08 Laboratory experiments based on the following-

1. Understanding Design Thinking and Problem Identification: Introduction to Design Thinking and Innovation Process, dentification of real-world problems from society, campus, healthcare, agriculture, education, environment, or industry, Stakeholder mapping and user identification, Conducting interviews, observations, and empathy mapping,
2. Define and Ideate: Problem statement formulation using "How Might We (HMW)" questions, Root cause analysis using Fishbone Diagram and 5-Why Technique, Brainstorming and idea generation

techniques, Idea screening using Impact–Feasibility Matrix or Decision Matrix, Storyboarding and user journey mapping.

3. Prototype Development: Low-fidelity prototype development (paper prototype, mock-up, sketch, wireframe, or physical model), Digital prototype creation using appropriate software/tools (e.g., Figma, Canva, Tinkercad, MIT App Inventor, or equivalent), Preparing value proposition and solution canvas, Prototype refinement based on peer feedback.
4. Testing and Project Presentation: User testing and feedback collection, Prototype modification and validation, Preparation of Business Model Canvas (basic introduction), Final team project presentation, demonstration, and documentation.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
-	1 * 22 NCH = 22 NCH	8 * 1 NCH = 8 NCH (Seminar, Case Study, Discussion, Internship)

Text Books:

1. *Developing Thinking Skills (The Way to Success)*, E. Balaguruswamy, 1st Edition, 2022, Khanna Publishing House
2. *Design Thinking for Engineering: A practical guide;* Iñigo Cuiñas, Manuel José Fernández Iglesias, 2023, Institution of Engineering and Technology
3. *Design Thinking For Strategic Innovation: What They Can't Teach You at Business or Design School,* Idris Mootee, 1st Edition, 2014, Adams Media
4. Chris Hackett, *The Big Book of Maker Skills: Tools & Techniques for Building Great Tech Projects,* Reprint Edition, 2018, Weldon Owen
5. Paul Horowitz, Winfield Hill, *The Art of Electronics,* 3rd Edition, 2015, Cambridge University Press

Reference Books:

1. Christian Müller-Roterberg; *Design Thinking For Dummies*, 1st Edition, 2020, For Dummies
2. *A Text Book of DESIGN THINKING For B.TECH. 4th Year, Semester-VII, Suitable For All The 4th Year B-Tech Students*
3. Simon Monk, *Programming Arduino: Getting Started with Sketches*, 2nd Edition, 2016, McGraw Hill TABH
4. Simon Monk, *Make Your Own PCBs with EAGLE: From Schematic Designs to Finished Boards,* 2014, McGraw Hill Education
5. Scott Chacon, Ben Straub, *Pro Git*, 2nd Edition, 2014, A Press
6. Chapman W.A.J, *Workshop Technology*, 5th Edition, 2002, CBS Publishers and distributors

Additional Reading:

1. https://www.aicte-india.org/sites/default/files/IDC/idealab/AICTE%20-%20IDEA%20LAB%20User%20Manual.pdf
2. https://vignaniit.edu.in/ideaLab.php

6.3 Detailed Syllabus of 2nd Semester

Paper I/Subject Name: Chemistry		Subject Code: CHY022C201
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objectives of the course are to s to acquaint the students with the basic phenomenon/concepts of chemistry, the student faces during course of their study in the industry and Engineering field and to understand the new developments and breakthroughs efficiently in engineering and technology

Prerequisites: Concepts of +2 level Chemistry

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the basic properties of chemical reactions	**BT 1**
CO 2	**Interpret** periodic properties such as ionization potential, electronegativity, oxidation states, electronegativity and bulk	**BT 2**
CO 3	**Experiment with** major chemical reactions that are used in the synthesis of molecules.	**BT 3**
CO 4	**Analyze** microscopic chemistry in terms of atomic and molecular orbitals and intermolecular	**BT 4**

Detailed Syllabus:

Module	Topics	Course Content	Periods
I.	**Atomic and Molecular Structure**	Schrodinger equation. Particle in a box solution and their applications for conjugated molecules and nanoparticles. Forms of the hydrogen atom wave functions and the plots of these functions to explore their spatial variations. Molecular orbitals of diatomic molecules and plots of the multicenter orbitals. Equations for atomic and molecular orbitals. Energy level diagrams of diatomic. Pi-molecular orbitals of butadiene and benzene and aromaticity. Crystal field theory and the energy level diagrams for transition metal ions and their magnetic properties. Band structure of solids and the role of doping on band structures.	**16**
II.	**Spectroscopic Techniques and Applications, Intermolecular Forces and Potential Energy Surfaces**	Principles of spectroscopy and selection rules. Electronic spectroscopy. Fluorescence and its applications in medicine. Vibrational and rotational spectroscopy of diatomic molecules. Applications. Nuclear magnetic resonance and magnetic resonance imaging, surface characterization techniques. Diffraction and scattering. Ionic, dipolar and van Der Waals interactions. Equations of state of real gases and critical phenomena. Potential energy surfaces of H3, H2F and HCN and trajectories on these surfaces.	**17**
III.	**Use of free**	Thermodynamic functions: energy, entropy and free energy. Estimations of entropy and free energies. Free energy and EMF. Cell potentials, the Nernst equation and applications. Acid base, oxidation	**16**

	Energy in Chemical Equilibria and Periodic Properties	reduction and solubility equilibria. Water chemistry. Corrosion. Use of free energy considerations in metallurgy through Ellingham diagrams. Effective nuclear charge, penetration of orbitals, variations of s, p, d and f orbital energies of atoms in the periodic table, electronic configurations, atomic and ionic sizes, ionization energies, electron affinity and electronegativity, polarizability, oxidation states, coordination numbers and geometries, hard soft acids and bases, molecular geometries	
IV.	**Stereochemistry, Organic Reactions and Synthesis of a Drug Molecule**	Representations of 3 dimensional structures, structural isomers and stereoisomers, configurations and symmetry and chirality, enantiomers, diastereomers, optical activity, absolute configurations and conformational analysis. Isomerism in transitional metal compounds Introduction to reactions involving substitution, addition, elimination, oxidation, reduction, cyclization and ring openings. Synthesis of a commonly used drug molecule	**17**
TOTAL			**66**

Chemistry Lab Syllabus

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Determination of surface tension and viscosity.
2. Thin layer chromatography.
3. Ion exchange column for removal of hardness of water.
4. Determination of chloride content of water.
5. Colligative properties using freezing point depression.
6. Determination of the rate constant of a reaction.
7. Determination of cell constant and conductance of solutions.
8. Potentiometry - determination of redox potentials and EMFs.
9. Synthesis of a polymer/drug.
10. Saponification/acid value of an oil.
11. Chemical analysis of a salt.
12. Lattice structures and packing of spheres.
13. Models of potential energy surfaces.
14. Chemical oscillations- Iodine clock reaction.
15. Determination of the partition coefficient of a substance between two immiscible liquids.
16. Adsorption of acetic acid by charcoal.
17. Use of the capillary viscosimeters to the demonstrate of the isoelectric point as the pH of minimum viscosity for gelatin sols and/or coagulation of the white part of egg

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books

1. *A Textbook of Physical Chemistry*, Negi A.S. and Anand S.C., 2nd Edition, 2007, New Age International
2. *Concise Inorganic Chemistry,* Lee J.D., 5th Edition, 2008, John Wiley and Sons Ltd.

Reference Books:

1. Atkins, P.W. and Paula, J. De,*Physical Chemistry,* 10th Edition, 2014, Oxford University Press
2. Huheey, J.E. Keiter, E.A. Keiter, R.L Medhi, O.K.,*Inorganic Chemistry Principles of Structure and Reactivity*, 4th Edition, 2006, Pearson Education

Additional Readings:

1. Organic Chemistry: Structure and Function by K. P. C. Volhardt and N. E. Schore, 5th Edition, http://bcs.whfreeman.com/vollhardtschore5e/default.asp
2. NPTEL Course on Chemistry - I by Prof. Mangala Sunder Krishnan, IITM

Paper II/Subject Name: Mathematics-II		**Subject Code: MAT022C202**
L-T-P-C – 3-0-1-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to teach the students Mathematics fundamentals necessary to formulate, solve and analyze engineering problems

Prerequisites: Concepts of Mathematics I

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** the methodologies used for solving various equations	**BT 1**
CO 2	**Understand** essential tool of matrices and linear algebra in a comprehensive manner	**BT 2**
CO 3	**Utilize** the essential tools in the field of applied sciences and related fields.	**BT 3**
CO 4	**Analyze** and **evaluate** the qualitative behavior of solutions of systems of differential equations and interpret in the context of an underlying model.	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Matrices** .	Linear Systems of Equations; Linear Independence; Rank of a Matrix; Determinant, Inverse of a matrix, rank-nullity theorem; System of linear equations; Symmetric, skew-symmetric and orthogonal matrices; Determinants; Eigenvalues and eigenvectors; Orthogonal transformation; Diagonalization of matrices; Cayley-Hamilton Theorem.	**15**

II	**Differential Equations**	Exact, linear and Bernoulli's equations. Equations not of first degree: equations solvable for p, equations solvable for y, equations solvable for x and Clairaut's type. Second order linear differential equations with variable coefficients: Euler-Cauchy equations, solution by variation of parameters; Power series solutions: Legendre's equations and Legendre polynomials, Frobenius method, Bessel's equation and Bessel's functions of the first kind and their properties.	**20**
III	**Complex Variable Differentiation**	Differentiation, Cauchy-Riemann equations, analytic functions, harmonic functions, finding harmonic conjugate; elementary analytic functions (exponential, trigonometric, logarithm) and their properties; Conformal mappings, Mobius transformations and their properties.	**15**
IV	**Complex Variable Integration**	Contour integrals, Cauchy-Goursat theorem (without proof), Cauchy Integral formula (without proof), Liouville's theorem and Maximum-Modulus theorem (without proof); Taylor's series, zeros of analytic functions, singularities, Laurent's series; Residues, Cauchy Residue theorem (without proof), Evaluation of definite integral involving sine and cosine, Evaluation of certain improper integrals using the Bromwich contour	**16**
TOTAL			**16**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A text book of Engineering Mathematics*, Bali N. P. and Narayan Iyenger N., 9th Edition, 2016, Laxmi Publication.
2. *Mathematical Methods for Physics and Engineering: A Comprehensive Guide,* K. F. Riley, M. P. Hobson, 3rd Edition, 2006, Cambridge University Press

Reference Books:

1. Grewal B. S., *Higher Engineering Mathematics,* 43rd Edition, 2014, Khanna Publishers.
2. Raisinghania M.D., *Ordinary and Partial Differential Equations*, 17th Edition, 2014, S. Chand and Co., New Delhi.
3. Narayna S., *A Text Book of Vector Calculus,* Revised Edition, 2009, S. Chand & Co., New Delhi.

Additional Readings:

1. https://mrcet.com/downloads/digital_notes/HS/R-18%20Mathematics-II.pdf
2. http://www.bosecuttack.in/studentcorner/LECTURE_NOTE.MATH2.2ND_SEM__1_.pdf
3. https://www.srividyaengg.ac.in/coursematerial/Iyear/111223.pdf

Paper III/Subject Name: Basic Environmental Engineering		Subject Code: CSE022C203
L-T-P-C – 3-0-0-3	Credit Units: 03	Scheme of Evaluation: T

Objective:

This course focuses on understanding various bio-potentials and their propagation, exploring different types of electrodes and their optimal placement for diverse recordings. It includes the design of bio-amplifiers for physiological recordings and examines measurement techniques for non-physiological parameters. Additionally, it aims to provide familiarity with biochemical measurement methods.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the structure and functions of ecosystems, the importance of biodiversity, and the principles of sustainable development including the UN SDGs.	**BT 1 & 2**
CO 2	**Apply** principles of water and wastewater treatment, solid waste management, and sustainable sludge disposal techniques in environmental engineering contexts.	**BT 3**
CO 3	**Apply** knowledge of air pollution, soil/groundwater contamination, and climate change to suggest appropriate mitigation and remediation strategies.	**BT 3**
CO 4	**Make** decisions using environmental modelling techniques, life cycle assessment (LCA), and risk assessment frameworks to evaluate environmental policies and standards such as ISO 14001.14001.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Ecosystems, Sustainability & Biodiversity**	Introduction to ecology and ecosystem structure and function - Sustainable development principles and UN Sustainable Development Goals (SDGs) - Biodiversity: value, threats, conservation strategies	**15**
II	**Water and Waste Water Treatment**	Principles of water, wastewater, Stormwater and industrial water treatment and management practices, Solid waste management: Sludge handling and sustainable disposal techniques	**10**
III	**Air Pollution**	Air pollutants, sources, transport, and fate, Soil and groundwater remediation: bioremediation, phytoremediation, emerging nanotechnologies, Climate change and global warming: causes and mitigation strategies	**15**

Modules	Topics	Course content	Hours
IV	**Environmental Modelling, Policy & Regulation**	Mass balance and life cycle assessment (LCA) in environmental systems, Environmental modelling approaches (numerical/software-based) , Environmental laws and policies: Indian and international (Air Act, Water Act, EPA, Solid Waste Rules), Risk assessment frameworks and ISO standards (e.g., ISO 14001)	**20**
TOTAL			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Biomedical Instrumentation and measurement,* Leslie Cromwell, 2nd Edition, 1990, Prentice Hall of India, New Delhi.
2. *Medical Instrumentation Application and Design,* John G. Webster, 4th Edition, 2009, John Wiley and sons, New York, (Units I, II & V)

Reference Books:

1. Myer Kutz, *Standard Handbook of Biomedical Engineering and Design*, 202, McGraw Hill Publisher
2. Khandpur R.S, *Handbook of Biomedical Instrumentation*, 3rd Edition, 2014, Tata McGraw-Hill, New Delhi, (Units II & IV)
3. Joseph J. Carr and John M. Brown, *Introduction to Biomedical Equipment Technology*, 4th Edition, 2004, Pearson Education.
4. R. Anandanatarajan, *Biomedical Instrumentation*, Kindle Edition, 2015, PHI Learning
5. M. Arumugam, *Biomedical Instrumentation*, 2003, Anuradha Agencies Publishers

Additional Readings:

1. https://www.studocu.com/in/document/aryabhatta-knowledge-university/btechit-btechcse/biology-notes-for-engineers/61016774
2. https://www.aminotes.com/2017/02/biology-for-engineers-module-1-cocepts.html
3. https://topperworld.in/b-tech-biology-notes/

Paper IV/Subject Name: Programming for Problem Solving-II		**Subject Code: CSE022C204**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students develop modular programming skills using functions, understand memory organization through pointers, implement user-defined data types, perform file processing in C.

Prerequisites: Concepts of Programming for Problem Solving-I

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Demonstrate** the concepts of functions, pointers, structures and files in C	**BT 2**
CO 2	**Develop** modular programs using functions and recursion.	**BT 3**
CO 3	**Analyze** and implement programs using pointers and dynamic memory allocation for efficient memory management.	**BT 4**
CO 4	**Design** programs using structures, unions, enumerated data types, and user-defined data types to model real-world problems.	**BT 6**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Concepts of Functions**	Function fundamentals, Function declaration, definition and calling, Function prototypes, Types of functions, Parameter passing in functions, call by value, Passing arrays to functions: idea of call by reference. Storage classes, Scope and lifetime of variables, Recursion, Recursion as a different way of solving problems. Example programs, such as Finding Factorial, Fibonacci series, Ackerman function etc, programming exercises, applications	**15**
II	**Pointers and Dynamic Memory Allocation**	Introduction to pointers, Pointer declaration and initialization, Pointer arithmetic, Pointers and arrays, Pointers and strings, Pointers as function arguments, Pointer to pointer, Dynamic memory allocation, malloc(), calloc(), realloc(), free(), Dangling and null pointers, programming exercises, applications	**15**
III	**User Defined Datatypes**	Structures, properties, advantages, initialization, working, Arrays of structures, Nested structures, Passing structures to functions, Pointer to structures, Self-referential structures (concept only), Unions, Enumerated data types (enum, typedef Bit fields (overview), programming exercises, applications	**15**
IV	**File Handling and Preprocessor Directives**	Introduction to files, File opening and closing, Text and binary files, File input/output functions, Random access files, Error handling in files, Command-line arguments, C preprocessor, #include, #define, Macros, Conditional compilation, Multi-file programming (overview), programming exercises, applications	**15**
TOTAL			**60**

Programming for Problem Solving-II Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Programs using user-defined functions
2. Recursive function applications

3. Programs demonstrating storage classes
4. Pointer operations
5. Pointer arithmetic
6. Pointers with arrays
7. Pointers with strings
8. Dynamic memory allocation
9. Structure operations
10. Arrays of structures
11. Nested structures
12. Union applications
13. File read/write operations
14. Random file access
15. Mini project involving modular programming and file handling

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	1*30 NCH = 30 NCH	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Computer Fundamentals and Programming in C,* Reema Thareja, 2nd Edition, 2016, Oxford University Press, Delhi.

Reference Books:

1. E Balaguruswamy, *Computing Fundamentals and C Programming*, 1st Edition, 2017, McGraw Hill.
2. Venugopal and Prasad, *Mastering C*, 2nd Edition, 2017, Tata McGraw Hill.
3. Yashawant Kanetkar,*Let us C,* 15th Edition, 2017, BPB.

Additional Readings:

3. https://mrcet.com/downloads/digital_notes/HS/Programming%20for%20Problem%20Solving.pdf
4. NPTEL course on Introduction to Programming in C by Prof. Satyadev Nandakumar, IIT, Kanpur

Paper V/Subject Name: Workshop Practices		**Subject Code: MEE022C215**
L-T-P-C – 0-0-4-2	**Credit Units: 02**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to provide exposure to the students with hands on experience on various basic engineering practices in Civil, Mechanical, Electrical and Electronics Engineering.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Label** the various techniques used under mechanical engineering	**BT 1**

CO 2	**Understand** the different manufacturing processes which are commonly employed in the industry	**BT 2**
CO 3	**Utilize** tools, instruments and techniques learnt to perform basic household chores in terms of house wiring, carpentry etc.	**BT 3**
CO 4	**Experiment** using the tools and techniques learnt for various purposes and decide on the best prospect.	**BT 4**

Detailed Syllabus:

Total Lab Hours for the semester = 40 (4 hours per week)

Minimum 10 Laboratory experiments based on the following-

The lecture sessions will be on the following topics:

- Manufacturing Methods- casting, forming, machining, joining, advanced manufacturing methods.
- CNC machining, Additive manufacturing.
- Fitting operations & power tools.
- Electrical & Electronics.
- Carpentry.
- Plastic moulding, glass cutting.
- Metal casting.
- Welding (arc welding & gas welding), brazing g topics:

And the lab sessions will on the topics:

- Machine shop
- Fitting shop
- Carpentry
- Electrical & Electronics
- Welding shop (Arc welding + Gas welding)
- Casting
- Smithy
- Plastic moulding & Glass Cutting

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
-	2*20 NCH = 40 NCH	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Elements of Workshop Technology,* Hajra Choudhury, S K, Hajra Choudhury, A K, 14th Edition, 2007, Mumbai Media Promoters
2. *Manufacturing Technology – I,* Gowri P. Hariharan and A. Suresh Babu, 2008, Pearson Education.

Reference Books:

1. Roy A. Lindberg, *Processes and Materials of Manufacture*", 4th Edition, 1998, Prentice Hall India,

Additional Readings:

1. http://mm-coep.vlabs.ac.in/LaserSpotWelding/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20Micromachining%20laboratory
2. http://fab-coep.vlabs.ac.in/exp7/Theory.html?domain=Mechanical%20Engineering&lab=Welcome%20to%20FAB%20laboratory

Paper VI/Subject Name: English for Technical Writing		**Subject Code: CEN022A201**
L-T-P-C – 2-0-0-2	**Credit Units: 02**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide learning environment to practice listening, speaking, reading and writing skills, to assist the students to carry on the tasks and activities through guided instructions and materials and to effectively integrate English language learning with employability skills and training.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the various forms of communication	**BT 1**
CO 2	**Understand** basic proficiency in English.	**BT 2**
CO 3	**Develop** reading and listening comprehension, writing and speaking skills.	**BT 3**
CO 4	**Analyze** the type of communication	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Vocabulary Building**	The concept of Word Formation, Root words from foreign languages and their use in English, Acquaintance with prefixes and suffixes from foreign languages in English to form derivatives, Synonyms, antonyms, and standard abbreviations.	**10**
II	**Basic Writing Skills**	Sentence Structures, Use of phrases and clauses in sentences, Importance of proper punctuation, creating coherence, organizing principles of paragraphs in documents, Techniques for writing precisely, Subject-verb agreement, Noun-pronoun agreement, Misplaced modifiers, Articles, Prepositions, Redundancies, Clichés	**10**

III	**Writing Practices**	Nature and Style of sensible Writing, Describing, Defining 1.3. Classifying, providing examples or evidence, Writing introduction and conclusion, Comprehension, Précis Writing, Essay Writing	**10**
IV	**Oral Communication**	Listening Comprehension, Pronunciation, Intonation, Stress and Rhythm, Common Everyday Situations: Conversations and Dialogues, Communication at Workplace, Interviews, Formal Presentations	**10**
		TOTAL	**40**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2*20 NCH = 40 NCH	-	20 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Effective Communication Skills.* Kul Bhushan Kumar, 2022, Khanna Book Publishing
2. *Practical English Usage,* Michael Swan. 1995, OUP

Reference Books:

1. F.T. Wood, *Remedial English Grammar.,* 2007, Macmillan.
2. William Zinsser, *On Writing Well*, 2001, Harper Resource Book.
3. Liz Hamp-Lyons and Ben Heasly, *Study Writing*, 2006,
4. Sanjay Kumar and PushpLata, *Communication Skills*, 2011, Oxford University Press.

Additional Readings:

1. AICTE's Prescribed Textbook: English (with Lab Manual), Khanna Book Publishing Co., https://khannabooks.com/index.php?route=product/product&path=99_105&product_id=480
2. NPTEL Course on English Language for Competitive Exams by Prof. by Aysha Iqbal, IIT, Madras
3. NPTEL Course on Technical English for Engineers by Prof. by Aysha Iqbal, IIT, Madras

Paper VII/Subject Name: Sports and Yoga		**Subject Code: CSE022C217**
L-T-P-C – 0-0-2-1	**Credit Units: 01**	**Scheme of Evaluation: P**

Objective:

The objectives of the course are to make the students understand the importance of sound health and fitness principles as they relate to better health, to expose the students to a variety of physical and yogic activities aimed at stimulating their continued inquiry about Yoga, physical education, health and fitness and to develop among students an appreciation of physical activity as a lifetime pursuit and a means to better health.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Choose** the best form of yoga/ exercise for them	**BT 1**
CO 2	**Understand** basic skills associated with yoga and physical activities including strength and flexibility, balance and coordination	**BT 2**
CO 3	**Experiment with** different forms of yoga to keep oneself physically fit and mentally strong	**BT 3**
CO 4	**Assess** current personal fitness levels	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Physical Education, Olympic Movement, Fitness, Wellness & Lifestyle**	Meaning & definition of Physical Education. Aims & Objectives of Physical Education. Changing trends in Physical Education, Ancient & Modern Olympics (Summer & Winter), Olympic Symbols, Ideals, Objectives & Values, Awards and Honours in the field of Sports in India (Dronacharya Award, Arjuna Award, Dhayanchand Award, Rajiv Gandhi Khel Ratna Award etc., Meaning & Importance of Physical Fitness & Wellness, Components of Physical fitness, o Components of Health-related fitness, Components of wellness, Preventing Health Threats through Lifestyle Change, Concept of Positive Lifestyle	**5**
II	**Anatomy & Physiology in Physical Education, Sports, Yoga & Postures**	Define Anatomy, Physiology & Its Importance, ffect of exercise on the functioning of Various Body Systems. (Circulatory System, Respiratory System, Neuro-Muscular System etc.), Meaning & Importance of Kinesiology & Biomechanics in Physical Edu. & Sports, o Newton's Law of Motion & its application in sports. o Friction and its effects in Sports, Meaning and Concept of Postures, Causes of Bad Posture. Advantages & disadvantages of weight training. Concept & advantages of Correct Posture. Common Postural Deformities – Knock Knee; Flat Foot; Round Shoulders; Lordosis, Kyphosis, Bow Legs and Scoliosis. Corrective Measures for Postural Deformities	**5**
III	**Yoga & Lifestyle**	Meaning & Importance of Yoga, Elements of Yoga, Asanas, Pranayama, Meditation & Yogic Kriyas, yoga for concentration & related Asanas (Sukhasana; Tadasana; Padmasana & Shashankasana) Relaxation Techniques for improving concentration - Yog-nidra, Asanas as preventive measures. Hypertension: Tadasana, Vajrasana, Pavan Muktasana, Ardha Chakrasana, Bhujangasana, Sharasana. Obesity: Procedure, Benefits & contraindications for Vajrasana, Hastasana, Trikonasana, Ardh Matsyendrasana. Back Pain: Tadasana, Ardh Matsyendrasana, Vakrasana, Shalabhasana, Bhujangasana. Diabetes: Procedure, Benefits & contraindications for Bhujangasana, Paschimottasana, Pavan Muktasana, Ardh Matsyendrasana. Asthema: Procedure, Benefits & contraindications for Sukhasana, Chakrasana, Gomukhasana, Parvatasana, Bhujangasana, Paschimottasana, Matsyasana.	**5**

IV	**Training, Planning and Psychology in Sports**	Meaning of Training, Warming up and limbering down, Skill, Technique & Style, Meaning and Objectives of Planning. Tournament – Knock-Out, League/Round Robin & Combination. Definition & Importance of Psychology in Physical Edu. & Sports, Define & Differentiate Between Growth & Development, Adolescent Problems & Their Management, Emotion: Concept, Type & Controlling of emotions, Meaning, Concept & Types of Aggressions in Sports. Psychological benefits of exercise. Anxiety & Fear and its effects on Sports Performance. Motivation, its type & techniques. Understanding Stress & Coping Strategies	**5**
TOTAL			**20**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
	1*20 NCH = 20 NCH	10 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Books:

1. *Modern Trends and Physical Education*, Ajmer Singh, Gill J.S, Bains J, 4th Edition, 2012, Kalyani Publishers

Reference Books:

1. B.K.S. Iyengar, *Light on Yoga*, 2006, Thorsons

6.4 Detailed Syllabus of 3rd Semester

Paper I/Subject Name: Discrete Mathematics		Subject Code: MAT022C301
L-T-P-C – 3-1-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students learn the concept of mathematical logic, sets, relations, and functions, generating functions and recurrence relations, Graph Theory for solving engineering related problems.

Prerequisites: Concepts of Mathematics I, II

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the concept of logic, sets, relations and functions to solve problems.	**BT 2**
CO 2	**Apply** the concepts learnt to solve computer science related problems.	**BT 3**
CO 3	**Analyze** and evaluate the solutions.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Sets, Relations, Functions & Algebraic Structures**	Operations and Laws of Sets, Binary, Relation, Partial Ordering Relation, Equivalence Relation, Functions, Inverse and Composite Function, Finite and infinite Sets, Countable and uncountable Sets, Poset, Lattice. The Well-Ordering Principle, The Division algorithm: Prime numbers, The Greatest Common Divisor, The least common multiple, Euclidean Algorithm, The Fundamental Theorem of Arithmetic, Congruence, Euler's phi function. Algebraic Structures with one Binary Operation, Semi Groups, Monoids, Groups, Congruence Relation and Quotient Structures, Free and Cyclic Monoids and Groups, Permutation Groups, Normal Subgroups, Algebraic Structures with two Binary Operation, Rings, Integral Domain and Fields.	**13**
II	**Graph Theory and Combinatorics**	Graphs and their properties, Degree, subgraphs, walks, paths and circuits, connected and disconnected graphs, Isomorphism, Eulerian and Hamiltonian graphs, Complete graphs, Bipartite graph, Trees, Properties of trees, Pendant vertex, Distance and Centers, Binary tree, Spanning trees, Planar graphs, Matrix representation of graphs, Chromatic number, Chromatic polynomial, Five colours theorem. Pigeon-hole principle, permutation and combination, Recurrence relations, Generating functions.	**20**
III	**Propositional Logic**	Proposition, connectives, tautology, contradiction, logical equivalence, normal forms-DNF, CNF, argument, Validity of	**20**

		argument, fallacy, Rules of Inference, Quantifiers. Boolean Algebra and Boolean Ring, Identities of Boolean Algebra, Duality, Representation of Boolean Function.	
IV	**Probability**	Definition of Probability, Classical, Relative Frequency, and Axiomatic Approaches, Basic Terminologies: Sample Space, Events, Outcomes, Types of Events: Mutually Exclusive, Exhaustive, Independent, Dependent, Addition Theorem of Probability, Conditional Probability, Multiplication Theorem of Probability, Bayes' Theorem and Its Applications, Discrete and Continuous Random Variables, Probability Mass Function (PMF) and Probability Density Function (PDF), Expectation, Variance, and Standard Deviation, Common Discrete Distributions, Joint and Marginal Probability, Independent and Dependent Events, Conditional Independence	**13**
		TOTAL	**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	2 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *A text book of Discrete Mathematics*, Sarkar S. K., Revised Edition, 2016, S Chand & Co Ltd.

Reference Books:

1. Deo N; *Graph Theory with applications to engineering and computer science*, New Edition, 2009, PHI Learning Private Limited.
2. Chandrasekaran N. and Umaparvathi, *Discrete Mathematics*, Eastern Economic Edition, 2013, PHI
3. *Discrete Mathematics and its Applications,* Rosen, K.H., 6th Edition, 2006, McGraw Hill.
4. Tremblay, J.P. and Manohar, R., *Discrete Mathematical Structures with Applications to Computer Science*, 35th Reprint, 2007, Tata McGraw Hill

Paper II/Subject Name: Data Structure & Algorithms		**Subject Code: CSE022C302**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students understand about the data structures, how to implement them in C, their advantages and drawbacks, & how they can be overcome.

Prerequisites: Concepts of Computer Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** various data structures used in programming.	**BT 1**

CO 2	**Understand** the basic constructs of data structure and its implementation.	**BT 2**
CO 3	**Utilise** the appropriate data structures to solve a given problem.	**BT 3**
CO 4	**Analyse** and **evaluate** the data structures used for problem solving	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course content	Hours
I	**Linear Data Structure-I**	a. Introduction: Why we need data structure? Concepts of data structures: Data and data structure, Abstract Data Type and Data Type. Algorithms and programs, basic idea of pseudo-code. Algorithm efficiency and analysis, time and space analysis of algorithms – order notations. b. Array : Different representations – row major, column major. Sparse matrix - its implementation and usage. Array representation of polynomials. c. Linked List: Singly linked list, circular linked list, doubly linked list, linked list representation of polynomial and applications.	**15**
II	**Linear Data Structure-I**	**a. Stack and Queue:** Stack and its implementations (using array, using linked list applications. Queue, circular queue, dequeuers. Implementation of queue- both linear and circular (using array, using linked list), applications. **b. Recursion:** Principles of recursion – use of stack, differences between recursion and iteration, tail recursion. Applications - The Tower of Hanoi, Eight Queens Puzzle.	**18**
III	**Non-Linear Data Structures**	**a. Trees:** Basic terminologies, forest, tree representation (using array, using linked list). Binary trees - binary tree traversal (pre-, in-, post- order), threaded binary tree (left, right, full) - non-recursive traversal algorithms using threaded binary tree, expression tree. Binary search tree- operations (creation, insertion, deletion, searching). Height balanced binary tree – AVL tree (insertion, deletion with examples only). **b. Graphs:** Graph definitions and concepts (directed/undirected graph, weighted/un-weighted edges, sub-graph, degree, cut-vertex/articulation point, pendant node, clique, complete graph, connected components – strongly connected component, weakly connected component, path, shortest path, isomorphism). Graph representations/storage implementations – adjacency matrix, adjacency list, adjacency multi-list. Graph traversal and connectivity – Depth-first search (DFS), Breadth-first search (BFS) – concepts of edges used in DFS and BFS (tree-edge, back-edge, cross-edge, and forward-edge), applications. Minimal spanning tree – Prim's algorithm (basic idea of greedy methods). B-Trees operation	**18**
IV	**Searching and Sorting**	**a. Sorting Algorithms:** Bubble sort and its optimizations, insertion sort, shell sort, selection sort, merge sort, quick sort, heap sort (concept of max heap, application – priority queue), radix sort. **b**. **Searching Algorithms:** Sequential search, binary search,	**15**

		interpolation search.	
TOTAL			**66**

Data Structures and Algorithms Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Some common programs of C as revision.
2. Programs on Arrays- Traversal, Insertion, Deletion, Polynomial Representation, etc.
3. Programs on Linked List- Creation Insertion, Deletion, Polynomial Representation, etc.
4. Programs on Stacks-Creation, Push Pop, Infix to Postfix Conversion, Evaluation.
5. Programs on Queues-Creation, Insertion, Deletion, etc.
6. Programs on Trees- Binary Tree Creation, Tree Traversal, BST
7. Programs on Searching- Linear Search, Binary Search
8. Programs on Sorting- Bubble Sort, Insertion Sort, Selection Sort, Quick Sort, Merge Sort, Heap Sort.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data Structures Using C*, Reema Thareja, 2nd Edition, 2014, Oxford University Press.

Reference Books:

1. Seymour Lipschutz, *Data Structures*, 1st Edition (reprint) 2017, McGraw Hill Education.
2. Narasimha Karumanchi, *Data Structures and Algorithms Made Easy: Data Structures and Algorithmic Puzzles*, 5th Edition, 2016, Careermonk Publications.
3. Horowitz, Sahni and Anderson-Freed, *Fundamentals of Data Structures in C*, 2nd Edition, 2008, Universities Press.
4. E. Balagurusamy, *Data Structures Using C*, 1st Edition, 2017, McGraw Hill Education.

Paper III/Subject Name: Computer Organization and Architecture		**Subject Code: CSE022C303**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain about the machine instructions and basic computer organization and I/O subsystems and pipelining processing

Prerequisites: Fundamental concepts of Digital Logic

Course Outcomes:

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the different hardware and its working in a Computer Systems in architectural level	**BT 1**
CO 2	**Demonstrate** computer architecture concepts related to design of modern processors, memories, and I/O	**BT 2**
CO 3	**Solve** problems related to computer Organization and Architecture	**BT 3**
CO 4	**Analyse** the performance of commercially available computers in architectural level.	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Content	Hours
I	**Basic organization of computers and machine instructions**	Block level description of the functional units as related to the execution of a program; Fetch, decode and execute cycle. Instruction set architectures, Assembly language programming, addressing modes, instruction cycles, registers and storage, addressing modes; discussions about RISC versus CISC architectures; Inside a CPU.	**15**
II	**Information representation**	Floating point representation (IEEE 754), computer arithmetic and their implementation; Fixed-Point Arithmetic: Addition, Subtraction, Multiplication and Division, Arithmetic Logic Units control and data path, data path components, design of ALU and data path, controller design; Hardwired and Microprogrammed Control	**18**
III	**Memory Technology**	Static and dynamic memory, Random Access and Serial Access Memories, Cache memory and Memory Hierarchy, Address Mapping, Cache updation schemes, Virtual memory and memory management unit.	**18**
IV	**I/O subsystems & Pipeline Processing**	Input-Output devices such as Disk, CD-ROM, Printer etc.; Interfacing with IO devices, keyboard and display interfaces; Basic concepts Bus Control, Read Write operations, Programmed IO, Concept of handshaking, Polled and Interrupt-driven I/O, DMA data transfer. Instruction and Arithmetic Pipeline, Pipeline hazards and their resolution, Parallel Processing.	**15**
		TOTAL	**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 22 NCH = 66 NCH	--	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Computer System and Architecture*, Moris Mano, 3rd Edition, 2007, PHI.
2. *Structured Computer Organization,* A. S. Tanenbaum, 5th Edition, 2009, Prentice Hall of India

Reference Books:

1. V. C. Hamacher, Z. G. Vranesic and S. G. Zaky, *Computer Organization,* 5th Edition, 20011, McGraw Hill.
2. J. L. Hennessy and D. A. Patterson, *Computer Architecture: A Quantitative Approach,* 5th Edition, 2011, Morgan Kaufmann.

Paper IV/Subject Name: Digital Logic and System Design		**Subject Code: CSE022C304**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students understand the simplification of Boolean expression and how to implement with various gates.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** the different types of circuits in system design	**BT 1**
CO 2	**Understand** the concepts of combinational and sequential circuit design	**BT 2**
CO 3	**Apply** the concepts learnt to design digital circuits.	**BT 3**
CO 4	**Analyse** the outputs produced and behaviour of the different circuits.	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course Content**	**Hours**
I	**Fundamental of Digital Electronics &Boolean algebra and its simplification**	Review of number system; Position number system – decimal, binary, octal and hexadecimal, number base conversion. Representation of negative binary numbers. Codes – BCD Gray, Excess -3 Digital signal, logic gates: AND, OR, NOT, NOR, EX-OR, EX-NOR Axioms and basic theorem of Boolean algebra. Truth table, logic function and their realization, standard representation (canonical forms) of logic gates-SOP and POS forms, MIN terms and MAX terms Simplification of logic function: K-map of 2, 3, 4 and 5 variables. Simplification of algebra and by map method. Don't care condition. Quine Mcluskey methods of simplification. Synthesis using AND, OR and INVERT and then to convert to NAND or NOR implementation	**18**

II	**Combinational logic circuit design**	Combinational logic circuit and buildings blocks. Binary adders and sub tractors. Carry look ahead adder. Encoders, Decoders, Multiplexers, Demultiplexers, Comparators, parity generators, etc. Realization of logic gates functions through decoders and multiplexers.	**15**
III	**Sequential circuits**	Flip flops: truth table and state table SR, JK, TD, race around condition, master slave conversion of flip-flops. Sequential shift register, sequence generator. Counter s: asynchronous and d Synchronous generators. Ring counter s and Johnson counter, up. Down counter modulo – N counter. Design of Synchronous sequential circuit.	**15**
IV	**Digital logic families and programmable logic devices**	Switching mode operation of PN junction, Bipolar and MOD device Bipolar families: RTL, DTL, DCTL, HTL, TTL, ECL, MOS, and CMOS logic families, Tristate logic. Gate properties fan in, fan out, propagation delay and power delay product. RAM and ROM their uses, SSI, MSI LSI and V LSI devices. Introduction to PLA, PAL TO FPGA and CPLDS, some commonly used digital ICs	**18**
TOTAL			**66**

Digital Logic and System Design Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 08 Laboratory experiments based on the following-

- To realize a transistorized AND Gate
- To realize a transistorized OR Gate
- To realize a transistorized NOT Gate
- To realize a transistorized NAND Gate
- To realize a transistorized NOR Gate
- To verify the truth tables of logic gates using ICs
- Realization of half and full adder
- Realization of half and full subtractor
- Realization of 2:1 and 1:2 DEMUX
- Realization of Encoder and Decoder

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Digital Logic & Computer Design*, M. Morris Mano, 1st Edition, 2016, Prentice Hall of India.
2. *Digital Principles and Applications*, P. Malvino and D. K. Leach, 8th Edition, 2014, Tata McGraw Hill.

Reference Books:

1. S. Salivahanan and S. Pravin Kumar, *Digital Logic Circuits*, 1st Edition, 2010, Vikas Publishing House.
2. Stephen Brown and Zvonko Vranesic, *Fundamentals of Digital Logic with VHDL Design*, 3rd Edition, 2017, McGraw Hill.
3. Sanjay Sharma, *Digital Electronics: Digital Logic Design*, 1st Edition, 2013, S K Kataria & Sons.
4. Pratima Manhas and Shaveta Thakral, *Digital Logic & Design*, 1st Edition, 2013, S K Kataria & Sons.
5. A Potton, *An Introduction to Digital Logic*, Imort Edition, 2013, Palgrave.

Paper V/Subject Name: OOP using C++		**Subject Code: CSE022C305**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students understand how C++ improves C with object-oriented features and to explain problem solving and programming skills in C++ with extensive programming projects.

Prerequisites: Fundamentals of Computer Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **understand** the basic concepts of OOP.	**BT 1 & 2**
CO 2	**Apply** the concepts learnt to write efficient programs in C++.	**BT 3**
CO 3	**Analyze** a problem and construct a C++ program that solves it.	**BT 4**
CO 4	**Assess** a C++ program and describe ways to improve it.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Introduction, Need, Characteristics, Difference between POP and OOP, Basic concepts of OOP, Features, Applications of OOP. Revision of topics like data types, keywords, identifiers, tokens, reference variables, different operators, conditional and loop control structures.	**15**
II	**Classes and Objects**	Definition of class, object, Difference between class and structure, class definitions, member functions, access specifiers. Objects Dynamic Creation and initialization, Passing and Returning objects, Object assignment and array of objects. Constructors Types, Destructors, Nesting member function, Private member function , Inline functions. Static class members, Function prototyping, Call by reference, Return by reference, Default Argument, Friend functions, this pointer.	**18**

III	**Inheritance and Polymorphism**	Types of Inheritance; Base and Derived classes, Syntax of derived classes, access to the base class; Types of Inheritance, Multiple inheritance, Virtual Base classes, Constructors and Destructors in Inheritance, Container classes, Abstract Classes. Polymorphism: Compile time(Early/Static binding), Overloading functions and operators,Overloading new and delete operators, Run time polymorphism(Late/Dynamic Binding), Virtual functions, Pure Virtual functions, Virtual Destructors, Review of Virtual base classes,	**18**
IV	**Templates, Exception and File Handling**	Templates–Uses, Generic classes, Class templates, Function templates, Advance templates. Examples. Exception handling-Advantages, Try catch and throw clauses, Examples,Manipulators, different examples of manipulators. Pointer types-uses; Dynamic memory allocation techniques, garbagecollection, Linked list, generic pointers; FilesOpen, Close, Read and Write; File attributes, File management	**15**
		TOTAL	**66**

OOP using C++ Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Write a C++ program to display "HELLO WORLD".
2. Write a C++ program that will ask the temperature in Fahrenheit and display in Celsius
3. Write a C++ program to print the following output using forloop.

   1
   2 2
   3 3 3
   4 4 4 4
4. Write a C++ program to reverse a number using do-whileloop
5. Write a C++ program to find out the factorial of a number using while loop
6. Write a C++ program to read an integer array and display it.
7. Write a C++ program to read a character array and display it.
8. Write a C++ program to find out the maximum of three number using if-elsestatement
9. Write a C++ program to implement the concept of static data member in class.
10. Write a C++ program to implement the concept of static function in class.
11. Write a C++ program using function with default argument.
12. Write a C++ program to illustrate the use of objects as function arguments (which performs the addition of time in the hour and minutes format)
13. Write a C++ program to illustrate the use of friend function.
14. Write a C++ program to illustrate how an object can be created (within a function) and returned to another function
15. Write a C++ program to illustrate the use of constructors and destructors.
16. Write a C++ program to illustrate the use of copy constructor.
17. Write a C++ program to implement single inheritance (private/public)
18. Write a C++ program to implement multilevel inheritance
19. Write a C++ program to implement multiple inheritances.
20. Write a C++ program to illustrate the use of virtual base class.
21. Write a C++ program to overload unary minus operator
22. Write a C++ program to overload binary „+" operator
23. Write a C++ program to illustrate how an operator can be overloaded using friend function.
24. Write a C++ program to illustrate the use of run time polymorphism.
25. Write a C++ program to swap two variable using function template

26. Write a C++ program to implement try(), catch(), throw()function.
27. Write a C++ program to implement this pointer
28. Write a C++ program to illustrate the use of pointers to derived objects
29. Write a C++ program to illustrate the use of virtual function
30. Write a C++ program to open and close a file using open(), close() function
31. Write a C++ program to illustrate the use of read(), write() function

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Object Oriented Programming With C++,* E. Balaguruswamy, 4th Edition, 2011, Tata McGraw Hill.
2. C++, *The Complete Reference,* Herbert Schildt, 4th Edition, 2017, McGraw Hill Education.

Reference Books:

1. Deital And Deital, *C++ How To Program,* 9th Edition, 2016, Pearson Education India.
2. R. Lafore, *Object Oriented Programming In Turbo C++*, 4th Edition, 2013, Galgotia, New Delhi

6.5 Detailed Syllabus of 4th Semester

Paper I/Subject Name: JAVA Programming		**Subject Code: CSE022C401**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the concepts and implementations of object-oriented programming using JAVA language.

Prerequisites: Basics of Procedural or Object-Oriented Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** basic structures, constructs, control structure and syntax of JAVA	**BT 1**
CO 2	**Understand** basic idea of object-oriented programming using JAVA	**BT 2**
CO 3	**Apply** the concepts of Java, multithreading and Exception handling to develop efficient and error free codes.	**BT 3**
CO 4	**Analyse** and **evaluate** JAVA programs for its efficiency	**BT 4 & 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	A look at procedure-oriented programming, Object-oriented paradigm, Basic concepts of object-oriented programming (OOP) (encapsulation, inheritance, polymorphism), How Java differs from C and C++, Applications of OOP. Overview of JAVA, Use of math functions, comments, Constructing a java program, Introduction of JVM, Command line argument, Data types, Variables: declaration, scope, Type Conversion and Type Casting, Constants, Operators, Evaluation of Expression, Precedence of Operators, Control statements: selection, iteration, and jump, Introduction to Socket Programming.	**15**
II	**Classes and Objects**	Class: definition and example, Declaring objects, Method overloading and overriding, Binding: concept of binding, static vs. dynamic binding, Using this and super keywords, Access Control, Inheritance: Extending a class, Final, Abstract classes, Constructors Arrays: one-dimensional and multi-dimensional, Strings: string processing functions	**15**
III	**Packages, Interfaces, Exception Handling**	Defining a package, accessing a package and using a package, Interfaces: multiple inheritance, Defining interfaces, implementing interfaces and extending interfaces. Exception handling fundamentals, Exception type: using try…catch, Multiple catch clauses, Throw and Throws Creating threads, Extending the thread class, Stopping and blocking a thread, Life cycle of thread, Threads methods, Thread exceptions, , JDBC,	**15**

IV	**Applets and Files**	Introduction: local and remote applets, How to write applets, Building applet code, Applet life cycle, Creating an executable applet I/O basics, concept of streams, Stream classes: byte stream classes, character stream classes, I/O exceptions, Creation of files, Random access files	**15**
Total			**60**

JAVA Programming Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Write a program in java that outputs your name in giant letters.
- Write a program in Java to find the day of the week of a given date.
- Write a program in Java called Grades Statistics, which reads in n grades (of int between 0 and 100, inclusive) and displays the average, minimum, maximum, and standard deviation.
- Write a program in Java to compute execution time by generating random numbers.
- Write a program in Java to implement the following:
    a. Tokenize the paragraph into single word.
    b. Find the number of word in a paragraph?
    c. Find the number of similar words from the input word.
    d. Find the number of occurrence of each word.
- Write a program in Java to implement some inheritance hierarchy.
- Design and implement an address book object that contains a person's name, home address and phone number, business address and phone number, and numbers for their fax machine, cellular phone, and pager. Write a program in Java to this test plan for the object and implement a driver [finally prepare a package].
- Write a program in Java to demonstrate the use of try, catch, finally throw and throws keywords and demonstrate the following points in the program.
    a. Multiple catch blocks.
    b. try-catch-finally combination.
    c. try-finally combination.
    d. Exception propagation among many methods.
    e. Use of getMessage(), printStackTrace() function of Throwable class.
    f. Nested try blocks
- Write a program that does the following.
    a. Prompts the user for an input file name through a dialog box.
    b. Prompts the user for an output file name through a dialog box.
    c. Copies the input file into the output file, subject to the removal of the space characters listed below from each line.
        i. The leading space characters
        ii. The trailing space characters
        iii. The space characters that are preceded by space characters
- Write a program in Java to design forms.
- Write a program in Java to design a student information form to enter data into the database.
- Write a program in Java to connect some form designed with the back-end database and test them by inserting some records.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 20 NCH = 60 NCH	2 * 15 NCH = 30 NCH	8 * 2 NCH = 16 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Programming with Java: A Primer*; Balagurusamy E., 3rd Edition, 2005, Tata McGraw-Hill, New Delhi
2. *Thinking in Java*, Eckel B., 4th Edition, 2006, PHI.

Reference Books:

1. Maurice N. et al, *Java Generics and Collections*, 1st Edition, 2006, O'REILLY Publication.
2. Booch G., Rumbaugh J. Jacobson I., *The Unified Modeling Language User Guide*, 2nd Edition, 2005, Pearson Education.
3. Schildt H., *The Complete Reference Java,* 7th Edition, 2007, Tata McGraw-Hill, New Delhi

Paper II/Subject Name: Database Management Systems		**Subject Code: CSE022C402**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to make the students learn about databases and the process of designing and constructing data models.

Prerequisites: C/C++, Concepts of Data Structures.

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **understand** the basic concepts and applications of database systems	**BT 1 & 2**
CO 2	**Apply** the basic concepts of MySql and write queries using it.	**BT 3**
CO3	**Analyze** the designed database for normalization.	**BT 4**
CO 4	**Evaluate** the process of transaction processing and concurrency control	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Basic Concepts**	Purpose of database systems-Components of DBMS –DBMS	**15**

		Architecture-Three Tier Architecture, and Data Independence-Data modelling -Entity Relationship Model and Diagram, Relational –Network-Hierarchical and object oriented models-Data Modelling using the Entity Relationship Model.	
II	**Structure of Relational Databases**	Relational databases –relational algebra-relational calculus, tuple and domain calculus. Data definition with SQL, insert, delete and update statements in SQL –views –data manipulation with SQL. assertions –triggers, Cursors	**18**
III	**Database Design**	Design guidelines–Relational database design –Integrity Constraints –Domain Constraints-Referential integrity –Functional Dependency-Normalization using Functional Dependencies, Normal forms based on primary keys-general definitions of Second and Third Normal Forms. Boyce-Codd Normal Form–Multi-valued Dependencies and Forth Normal Form –Join Dependencies and Fifth Normal Form –Pitfalls in Relational Database Design, Properties of Relational Decomposition, Dependency Preserving Property, Lossless Non-Additive Join Property, Testing Relational Decompositions for non-additive and dependency preserving properties.	**18**
IV	**Introduction to Transaction and Query Processing**	Transaction and System Concepts-Desirable properties of Transactions-Schedules and Recoverability-Serializability of Schedules -Concurrency Control–Data Storage Indexing and Query processing and Optimization MySQL case study: The basic structure of the MySQL system database structure and its manipulation in MySQL -storage organization in MySQL -Programming in PL/SQL-Cursor in PL/SQL	**15**
		TOTAL	**66**

Database Management Systems Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Programs to understand the functionality and limitations of file system.
2. Consider the following relational schema
   Employee (Emp_no, Name, Salary, design, dept_id, DOJ)
   Department (Dept_id, DName, loc, DOE)
   a. Display the name of the employees working in marketing dept.
   b. Display the details of the employee joined in the month of July.
   c. Display the details of the employee who gets maximum salary.
   d. Count the no of employees in each department
3. Consider the following relational schema
   Student (Rollno, Name, Address, DOB, C_id)
   Course (C_id, Cname, Dur, Fees)
   a. Display rollno,name,cname,fees of each student
   b. Count the no of students in each course
4. Consider the following relational schema
   Books(book_id,b_name,author,purchase_date,cost)
   Members(member_id,m_name,address,phone,birthdate)

Issue_return(book_id,member_id,issue_date,return_date)

a. Find the author of the books that have not been issued.
b. Display the member_id and no of books issued to that (Assume that if a book in Issue_Return relation does not have a return_date then it is issued)
c. Find the book that has been issued the minimum no of times.
d. Display the names and author of the books that have been issued at any time to a member whose name begins with "Ra".
e. Display the name and Cost of those books that have been issued to any member whose date of birth is less than 01-01-1989 but not been issued to any member having the birth date equal to or greater than 01-01-1989.

5. Consider the following relational schema

Student(name,phone,dob,s_id)
Course(c_id,cname,credit,teacher_id)
Result(s_id,c_id,mark)

a. Find the name of the students whose results are not declared in any course
b. Find the teachers who are teaching more than one course
c. Display the name and marks of those students who were born before 1-1-1989 and score more than 80 marks in any course
d. Find the details of students securing pass marks in more than 3 course
e. Find the total no of credits earned by a students whose id is 10.
f. Find name of the students who got maximum overall marks.
g. Display the name and marks of those students who scored more than 80 marks in any subject.
h. Find the details of the students securing less than 30 marks in more than 3 subjects.

6. Consider the following relational schema

Customer(C_id, Name , Address)
Item(i_code , Name , Price)
Purchase (P_id ,C_id , I_code, qty , pdate)

a. Find the name of the customer who has done maximum purchase.
b. Display the name of the item that has been purchased maximum no of times in the month of Feb.
c. Display the name of the customer who didn't purchase any item.

7. Create three triggers (insert, delete and update) on emp table so that:
   a. Whenever a new record is inserted then the emp_id and date of insertion is stored in another table called new_rec.
   b. Whenever a record is deleted the emp_id and date of deletion is stored in another table called old_rec.
   c. Whenever employee's salary is updated the emp_id , old salary and updated salary is stored in another table called update_info.
8. Write a procedure to accept a emp_id and display the employee details.
9. Write a procedure to accept a emp_id and return the employee salary.
10. Given,

Emp(emp_no,name,salary,supervisor_no,dept_code)
Dept(dept_code, dept_name)

a. employees who get more salary than their supervisor
b. Department name and total number of employees in each Department.
c. Name and department of employee(s) who earn maximum salary.

11. Programs on Views and Cursors

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Fundamentals of Database System,* Elmasri and Navathe, 7thEdition, 2016, Pearson Education Asia
2. *Database System Concepts*, Henry F Korth, Abraham Silberschatz, 6th Edition, 2013, Mc Graw Hill.
3. *DataBase Management System*, Paneerselvam, 2nd Edition, 2011, PHI Learning

Reference Books:

1. C. J. Date*, An Introduction to Database Systems,* 8th Edition, 2003, Pearson Education Asia
2. Bibin C. Desai, *An Introduction to Database Systems*, Revised Edition, 2012, Galgotia Publications

Paper III/Subject Name: Formal Language and Automata Theory		**Subject Code: CSE022C403**
L-T-P-C – 3-1-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to impart knowledge on regular grammars, regular expressions and to teach about the basics of parsing and ambiguity.

Prerequisites: Fundamentals of Set Theory

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**List** and **understand** the utility and importance of Automata Theory as the basis of all computer science languages design	**BT 1 & 2**
CO 2	**Construct** minimized sample automata and grammars of context free languages	**BT 3**
CO3	**Analyze** the power and limitation of a computer and solve the problems using formal language	**BT 4**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Basics of Strings and Alphabets, DFA, transition graphs, regular languages, non-deterministic FA, equivalence of DFA and NDFA	**15**
II	**Grammars**	Regular grammars, regular expressions, equivalence between regular languages, properties of regular languages, pumping lemma.	**18**
III	**Deterministic and Non-Deterministic PDA**	Leftmost and rightmost derivation, parsing and ambiguity, ambiguity in grammar and languages, normal forms. NDPDA, DPDA, context free languages and PDA, comparison of deterministic and non-deterministic versions, closure properties, pumping lemma for CFL	**18**
IV	**Turing Machine**	Turing Machines, variations, halting problem, PCP Chomsky Hierarchy Manipulators, different examples of manipulators; Pointer types- uses; Dynamic memory allocation techniques - garbage collection, Linked list, generic pointers; Files-	**15**

		Open, Close, Read and Write; File attributes, File management	
			TOTAL **66**

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *An Introduction to Formal Languages and Automata*, Peter Linz, 3rd Edition, 2010, Narosa Publishers

Reference Books:

1. J. E. Hopercroft and J. D. Ullman,*Introduction to Automata Theory, Languages & Computation*, 3rd Edition, 2006, Narosa
2. J. C. Martin, *Introduction to Languages and The Theory of Computation,* 3rd Edition, 2009, McGraw Hill International Edition.

Paper IV/Subject Name: Microprocessor		**Subject Code: CSE022C404**
L-T-P-C – 2-0-2-3	**Credit Units: 03**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to teach about the architecture, design aspects of I/O of and Memory Interfacing circuits of 8086 microprocessor and 8051 microcontroller.

Prerequisites: Concepts of Digital Logic and Operating Systems

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the basic concepts of microprocessor and microcontroller.	**BT 2**
CO 2	**Apply** the concepts learnt to design and implement programs on 8086 microprocessors, I/O and Memory Interfacing circuits and 8051 microcontroller-based systems.	**BT 3**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**

I	**8086 Microprocessor**	Introduction to 8086 – Microprocessor architecture – Addressing modes - Instruction set and assembler directives – Assembly language programming – Modular Programming - Linking and Relocation - Stacks - Procedures – Macros – Interrupts and interrupt service routines – Byte and String Manipulation.	**11**
II	**8086 System Bus Architecture**	8086 signals – Basic configurations – System bus timing – System design using 8086 – IO programming – Introduction to Multiprogramming – System Bus Structure - Multiprocessor configurations – Coprocessor, Closely coupled and loosely Coupled configurations – Introduction to advanced processors	**11**
III	**I/O Interfacing**	Memory Interfacing and I/O interfacing - Parallel communication interface – Serial communication interface – D/A and A/D Interface - Timer – Keyboard /display controller – Interrupt controller – DMA controller – Programming and applications Case studies: Traffic Light control, LED display , LCD display, Keyboard display interface and Alarm Controller.	**11**
IV	**8051 Microcontroller**	Architecture of 8051 – Special Function Registers(SFRs) - I/O Pins Ports and Circuits - Instruction set - Addressing modes - Assembly language programming. Programming 8051 Timers - Serial Port Programming - Interrupts Programming – LCD & Keyboard Interfacing - ADC, DAC & Sensor Interfacing - External Memory Interface- Stepper Motor and Waveform generation.	**11**
TOTAL			**44**

Microprocessor Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

EXPERIMENT NO.	AIM OF THE EXPERIMENT	HOURS
1	Introduction to MASM and TASM. Assembling and executing the programs.	3
2	Programs involving data transfer instructions a) Byte and word data transfer in different addressing mode b) Block move with overlap and without overlap c) Block exchange	3
3	Programs involving arithmetic and logic operation on signed and unsigned multi byte numbers a) 16-bit addition and subtraction b) 32 addition and subtraction c) 16-bit multiplication of signed and unsigned numbers d) 8-bit division of signed and unsigned numbers e) 16-bit division of signed and unsigned numbers	3

4	Code conversions a) Converting BCD into ASCII b) Binary to BCD c) BCD to binary	3
5	Program involving string manipulation a) String reversal b) Comparison of two strings c) Program to search for a character in a string d) Program to check for a palindrome	3
6	Programs involving branch / loop instructions a) Program to find largest and smallest in a series. b) Program to sort the numbers in ascending/ descending order. c) Addition of n numbers	3
	Total	**18**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
2* 22 NCH = 44 NCH	1 * 15 NCH = 30 NCH	8 * 2 NCH = 26 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Microcomputer Systems: The 8086 / 8088 Family - Architecture, Programming and Design*, Yu-Cheng Liu, Glenn A.Gibson, 2nd Edition, 2007, Prentice Hall of India,
2. *The 8051 Microcontroller and Embedded Systems: Using Assembly and C*, Mohamed Ali Mazidi, Janice Gillispie Mazidi, Rolin McKinlay, 2ndEdition, 2011, Pearson Education.

Reference Books:

1. Doughlas V.Hall, *Microprocessors and Interfacing, Programming and Hardware*, 2nd Edition, 2012, TMH

Paper V/Subject Name: Programming with Python		**Subject Code: CSE022G405**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the students about Programming with Python and use it to solve real world problems.

Prerequisites: Fundamentals of Computers

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** semantics and syntax of Typescripts.	**BT 1**

CO 2	**Understand** static types and know how to port untyped JavaScript	**BT 2**
CO 3	**Apply** the concepts learnt to create Single Page Web Applications (SPA) using React, Typescript and Tailwind CSS.	**BT 3**
CO 4	**Inspect** different elements of front-end development	**BT 4**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction to Python**	Introduction to Python Programming: Python interpreter/shell, indentation; identifiers and keywords; literals, numbers, and strings; operators (arithmetic operator, relational operator, Boolean operator, assignment, operator, ternary operator and bitwise operator) and expressions	**18**
II	**Programming With Python**	Input and output statements, defining functions, control statements (conditional statements, loop control statements, break, continue and pass, exit function.), default arguments,	**18**
III	**Python Functions and Strings**	Python Functions, Python Lambda, Python Arrays, Python Classes/Object, Inheritance, Iterator, Polymorphism, Scope, Modules, Dates, Maths, JSON,RegEx, PIP, User Input , Strings	**15**
IV	**Python Modules**	Introduction to Numpy, Pandas,SciPy, Dijango	**15**
Total			**66**

Introduction to Python Programming Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. **Hello World Program:**
   - Write a simple Python program to print "Hello, World!" to the console.
2. **Variable Declaration and Printing:**
   - Practice declaring variables of different types (int, float, string) and printing their values.
3. **Basic Arithmetic Operations:**
   - Write Python scripts to perform basic arithmetic operations such as addition, subtraction, multiplication, and division.
4. **Conditional Statements:**
   - Create programs using if-else statements to perform tasks based on certain conditions.
5. **Loops (for and while):**
   - Practice writing for and while loops to iterate over sequences or execute code repeatedly.
6. **Lists and List Operations:**
   - Explore lists in Python and perform operations like appending, removing, and accessing elements.

   Level: Intermediate
7. **Functions:**

- Define and call functions to encapsulate reusable code blocks. Practice passing arguments and returning values from functions.

8. **String Manipulation:**
    - Work on tasks involving string manipulation, such as concatenation, slicing, and searching.
9. **File Handling:**
    - Write Python scripts to read from and write to files. Practice handling exceptions during file operations.
10. **Dictionaries and Sets:**
    - Experiment with dictionaries and sets in Python. Perform operations like adding, removing, and accessing elements in dictionaries and sets.
11. **Object-Oriented Programming (OOP) Concepts:**
    - Introduce students to OOP concepts like classes, objects, inheritance, and polymorphism. Have them implement simple classes and explore inheritance hierarchies.
12. **Exception Handling:**
    - Practice handling exceptions using try-except blocks to gracefully manage errors in Python programs.
13. **Data Structures and Algorithms:**
    - Implement common data structures (e.g., stacks, queues, linked lists) and algorithms (e.g., sorting, searching) using Python.
14. **Regular Expressions:**
    - Introduce regular expressions and their usage in Python for pattern matching and text processing tasks.
15. **Modules and Packages:**
    - Explore the concept of modules and packages in Python. Have students create their own modules and packages and import them into other scripts.
16. **GUI Programming with Tkinter:**
    - Introduce GUI programming using Tkinter. Have students create simple graphical user interfaces (GUIs) for basic applications.

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*22 NCH = 66 NCH	--	3*8 = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Project)

Textbooks:

1. Guttag, J.V. (2016). Introduction to computation and programming using Python. 2nd edition. MIT Press.

Reference Books:

1. Kamthane, A. N., & Kamthane, A.A. (2017) Programming and Problem Solving with Python, McGraw Hill Education.
2. Liang, Y. D. (2013). Introduction to Programming using Python. Pearson Education.

6.6 Detailed Syllabus of 5th Semester

Paper I/Subject Name: Operating Systems		Subject Code: CSE022C501
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make the students understand the fundamental concepts and design of operating systems, apply principles of concurrency, synchronization, and deadlock in operating systems.

Prerequisites: Computer Programming, Computer Architecture & Organization

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **illustrate** the role and responsibilities of an operating system, system calls, kernel vs. user mode, etc.	**BT 1 & 2**
CO 2	**Apply** process scheduling algorithms (FCFS, SJF, RR, etc.) to compute performance metrics.	**BT 3**
CO 3	**Analyze** synchronization problems and solutions like semaphores, monitors, and deadlock.	**BT 4**
CO 4	**Evaluate** different memory management strategies and their impact on performance.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Operating Systems Overview**	Introduction and history of Operating systems, structure and operations; processes and files. **Computer System Overvie**w- Basic Elements, Instruction Execution, Interrupts Memory Hierarchy, Cache Memory, Direct Memory Access, Multiprocessor and Multicore Organization. **Operating system overview**-objectives and functions, Evolution of Operating System.- Computer System Organization- Operating System Structure and Operations- System Calls, System Programs, OS Generation and System Boot	**15**
II	**Process Management And Concurrency Control**	**Processes**-Process Concept, Process Scheduling, Operations on Processes, Interprocess Communication; Threads- Overview, Multicore Programming, Multithreading Models; Thread and SMP Management. **Process Synchronization** – Critical Section Problem, Mutex Locks, Semaphores, Monitors; CPU Scheduling and scheduling algorithms. **Deadlocks**- Shared resources, resource allocation and scheduling, resource graph models, deadlock detection, deadlock avoidance, deadlock prevention algorithms	**18**

III	**Storage Management**	Memory Management requirements, Memory partitioning: Fixed and Variable Partitioning, Memory Allocation: Allocation Strategies (First Fit, Best Fit, and Worst Fit), Fragmentation, Swapping, and Paging. Segmentation, Demand paging, Virtual Memory: Concepts, management of VM, Page Replacement Policies (FIFO, LRU, Optimal, Other Strategies), Thrashing. 32 and 64 bit architecture Examples; Allocating Kernel Memory, OS Examples	**18**
IV	**I/O and File Systems**	I/O Devices, Organization of I/O functions, Operating System Design issues, I/O Buffering, Overview of mass storage structure- disks and tapes. Disk structure – accessing disks, Swap Space. Disk Scheduling (FCFS, SCAN, C-SCAN, SSTF), RAID, Disk Cache. Disk Protection– Goals, Principles, Domain. **File System Interface**: File Concepts – Attributes – operations – types – structure – access methods. File system mounting. Protection. File system implementation. Directory implementation – allocation methods. Free space Management.	**15**
		TOTAL	**66**

Operating Systems Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. Basic Linux Commands and Overview.
2. Write Shell Script for followings
    - To find the global complete path for any file.
    - To broadcast a message to a specified user or a group of users logged on any terminal.
    - To copy the file system from two directories to a new directory in such a way that only the latest file is copied in case there are common files in both the directories.
    - To compare identically named files in two different directories and if they are same, copy one of them in a third directory
    - To delete zero sized files from a given directory (and all its sub- directories).
    - To display the name of those files (in the given directory) which are having multiple links.
    - To display the name of all executable files in the given directory.
    - Write a script to display the date, time and a welcome message (like Good Morning etc.). The time should be displayed with "a.m." or "p.m." and not in 24 hours notation.
    - Write a script to display the directory in the descending order of the size of each file.
3. Implementation of System Calls.
4. Implementation of FCFS (First Come First Serve) CPU Scheduling.
5. Implementation of SJF (Shortest Job First) CPU Scheduling.
6. Implementation of Round Robin (RR) CPU Scheduling.
7. Implementation of Priority CPU Scheduling Algorithm.
8. Implementation of FIFO Replacement Algorithm.
9. Implementation of Optimal Page Replacement Algorithm.
10. Implementation of LRU Page Replacement Algorithm by Stack method
11. Implement the producer-consumer problem using threads.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Operating System Concepts,* Abraham Silberschatz, Peter Baer Galvin and Greg Gagne, 9th Edition, 2012, John Wiley and Sons Inc.

Reference Books:

1. William Stallings, *Operating Systems – Internals and Design Principles*, 7th Edition, 2011, Prentice Hall.
2. Andrew S. Tanenbaum, *Modern Operating Systems*, 2nd Edition, 2001, Addison Wesley.
3. D M Dhamdhere, *Operating Systems: A Concept-Based Approach*, 2nd Edition, 2007, Tata McGraw-Hill Education.

Paper II/Subject Name: Introduction to Data Science		**Subject Code: CSE022C502**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:
The objective of this course is to introduce students to data science fundamentals, including data collection, preprocessing, exploratory analysis, statistical modeling, and basic machine learning techniques. The focus will be on structured and semi-structured data before transitioning to big data analytics in the next course.

Prerequisites: Basic computer programming, fundamental mathematics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	Understand the fundamental concepts and scope of Data Science	**BT 1 & 2**
CO 2	Perform data collection, cleaning, and preprocessing for structured and semi-structured data	**BT 3**
CO 3	Conduct exploratory data analysis (EDA) and apply statistical techniques for insight generation	**BT 4**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction to Data Science & Data Types**	Overview of Data Science, Key applications in real-world industries, Structured vs Semi-Structured vs Unstructured data, Data Science Workflow, Tools & Technologies (Python/R, Jupyter, SQL, Excel, BI Tools)	**12**
II	**Data Collection, Preprocessing & Wrangling**	Data collection techniques (APIs, Web Scraping, Databases), Handling missing values, Outlier detection, Encoding categorical variables, Data Transformation (Scaling, Normalization, Feature Engineering)	**12**
III	**Exploratory Data Analysis (EDA) & Statistical Techniques**	Descriptive Statistics, Data Visualization (Histograms, Box plots, Pair plots, Heatmaps), Correlation & Covariance, Hypothesis Testing (t-test, ANOVA, Chi-Square test), Feature Selection	**12**
IV	**Machine Learning for Structured Data**	Introduction to Supervised & Unsupervised Learning, Linear & Logistic Regression, Decision Trees, Model Evaluation (Accuracy, Precision, Recall, F1-Score, RMSE), Preparing for Big Data Analytics (Data Pipeline Concepts)	**12**
TOTAL			**48**

Introduction to Data Science Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Installation and setup of Anaconda, Jupyter Notebook, and Google Colab.
- Introduction to NumPy arrays and mathematical operations.
- Data manipulation using Pandas DataFrames.
- Reading and writing data from CSV, Excel, and JSON files
- Collecting data from public datasets and APIs.
- Handling missing values using deletion and imputation methods.
- Detecting and treating outliers.
- Data transformation, normalization, and standardization.
- Encoding categorical variables and feature engineering.
- Descriptive statistical analysis using Pandas.
- Data visualization using Matplotlib and Seaborn.
- Histograms, box plots, scatter plots, and pair plots.
- Correlation analysis and heatmap generation.
- Exploratory analysis on a real-world dataset.
- Implementation of t-test and Chi-Square test.
- ANOVA for comparing multiple groups.
- Feature selection using statistical methods.
- Linear Regression for prediction problems.
- Logistic Regression for classification problems.
- Decision Tree classification and regression.
- Model evaluation using Accuracy, Precision, Recall, F1-Score, and RMSE.
- Mini-project: End-to-end data science workflow on a real-world dataset.

National Credit Hours		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3 * 15 NCH = 45 NCH	1 * 30 NCH = 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Textbook:

1. *Doing Data Science,* Cathy O'Neil & Rachel Schutt, 1st Edition, 2013, O' Reily

Reference Books:

1. Foster Provost & Tom Fawcett, *Data Science for Business,* 1st Edition, 2013, O' Reily
2. Wes McKinney, *Python for Data Analysis,* 2nd Edition, 2017, O' Reily

Paper III/Subject Name: Design and Analysis of Algorithms		**Subject Code: CSE022C503**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to enable the students analyze performance of algorithms and solve problems using algorithm design methods such as the greedy method, divide and conquer, dynamic programming, backtracking and branch and bound.

Prerequisites: Concepts of Data Structures and Basic Mathematics

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Examine** and **understand** the performance of algorithm.	**BT 1 & 2**
CO 2	**Apply** different designing methods for development of algorithms to realistic problems, such as divide and conquer, greedy and etc.	**BT 3**
CO 3	**Analyze** and **evaluate** algorithms to improve their efficiency.	**BT 4 & 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction and Divide and Conquer**	Algorithm, Psuedo code for expressing algorithms, Performance Analysis-Space complexity, Time complexity, Asymptotic Notation- Big oh notation, Omega notation, Theta notation and Little oh notation, Probabilistic analysis, Amortized analysis. Master's Theorem	**15**

II	**Searching and Traversal Techniques**	Efficient non - recursive binary tree traversal algorithm, Disjoint set operations, union and find algorithms, Spanning trees, Graph traversals - Breadth first search and Depth first search, AND / OR graphs, game trees, Connected Components, Bi - connected components. Disjoint Sets- disjoint set operations, union and find algorithms, spanning trees, connected components and bi-connected components.	**18**
III	**Types of Problem Solving Techniques**	Greedy Method: General method, applications - Job sequencing with dead lines, 0/1 knapsack problem, Minimum cost spanning trees, Single source shortest path problem. Dynamic Programming: General method, applications-Matrix chain multiplication, Optimal binary search trees, 0/1 Knapsack problem, All pairs shortest path problem,Travelling sales person problem, Reliability design. Backtracking: General method, applications-n-queen problem, sum of subsets problem, graph coloring, Hamiltonian cycles. Branch and Bound: General method, applications - Travelling sales person problem,0/1 knapsack problem- LC Branch and Bound solution, FIFO Branch and Bound solution	**18**
IV	**NP-Hard and NP-Complete Problems**	NP-completeness – Polynomial time verification – Theory of reducibility – Circuit satisfiability - NP-completeness proofs – NP-complete problems: Vertex cover, Hamiltonian cycle and Traveling Salesman problems – Approximation Algorithms – Approximation algorithms to vertex-cover and traveling salesman problems.	**15**
TOTAL			**66**

Design and Analysis of Algorithms Lab

Detailed Syllabus:

Total Lab Hours for the semester = 48 (4 hours per week)

Minimum 20 Laboratory experiments based on the following-

- Implementation and analysis of iterative and recursive algorithms.
- Implementation of searching algorithms: Linear Search, Binary Search
- Implementation of sorting algorithms: Bubble Sort, Selection Sort, Insertion Sort
- Implementation of efficient sorting algorithms: Merge Sort, Quick Sort, Heap Sort
- Implementation of Strassen's method
- Implementation of programs using dynamic programming.
- Implementation of programs using greedy technique.
- Implementation of programs using backtracking technique.
- Implementation of programs using Branch and Bound technique.
- Implementation of graph-related algorithms.
- Implementation of programs using randomized technique.

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**

3* 15 NCH = 45 NCH	1*30 NCH = 30 NCH	45 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Introduction to Algorithms,* T. H. Cormen, C. E. Leiserson, R. L. Rivest, 3rd Edition, 2009, The MIT Press, Cambridge, Massachusetts.

Reference Books:

1. Aho, Hopcroft & Ullman, *The Design and Analysis of Algorithms,* 1974,Addison- Wesley
2. Horowitz & Sahani, *Fundamentals of Algorithms,* 2nd Edition, 2009, Galgotia Publications

Paper IV/Subject Name: Data Communication		**Subject Code: CSE022C504**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach about the fundamental principles and models of data communication, apply knowledge of modulation, multiplexing, and switching techniques, etc.

Prerequisites: Concepts of Computer Fundamentals, Digital Logic and Design.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the basic components and types of data communication systems and data transmission concepts.	**BT 1 & 2**
CO 2	**Apply** error detection and correction techniques such as CRC and Hamming code.	**BT 3**
CO 3	**Analyze** data link layer protocols such as HDLC, Stop-and-Wait, and Sliding Window	**BT 4**
CO 4	**Evaluate** performance issues in flow control, multiplexing, and switching techniques.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Introduction to Computer network, Networks: classification and components, Layered architecture of a network software, OSI and TCP/IP model. Data Transmission: Communication model- Simplex, half duplex and full duplex transmission - Periodic Analog signals: Sine wave, phase, wavelength, time and frequency domain, bandwidth – Digital Signals; Digital data Transmission:- Analog & Digital data, Analog & Digital signals, Analog &Digital transmission – Transmission Impairments: Attenuation, Delay distortion, Noise – Channel capacity:	**18**

II	**Signal Analysis**	Introduction to Signal and its Classification. System and its Basic Properties, Spectral Analysis of a Signal; Signal Bandwidth. Transmission media - Guided Transmission Media: Twisted pair, Coaxial cable, optical fiber, Wireless Transmission, Terrestrial microwave, Satellite microwave. Wireless Propagation: Ground wave propagation, Sky Wave propagation, LoS Propagation.	**15**
III	**Data Encoding and Multiplexing**	Baseband Communication: Data Encoding and Modulation, Analog Modulation: AM, FM and PM, Pulse Modulation System: PAM and PWM. Digital Modulation: ASK, FSK, PSK and QAM Multiplexing and its Application: Frequency Division Multiplexing, Wavelength Division Multiplexing, Time Division Multiplexing, Spread Spectrum.	**15**
IV	**Switching and Information Theory and Coding**	Switching: Switching and its Application, Circuit Switching and Packet Switching, Datagram Switching and Virtual Circuit SwitchingX.25, Frame Relay and ATM. Introduction to Information Theory and Average Information Source Coding: Huffman Coding, Error Detection and Correction Codes, Hamming Distance, Linear Block Coding, Cyclic Codes, CRC, Convolution Codes	**18**
TOTAL			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	--	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data Communications and Networking*, Forouzan B. A., 4th Edition, 2007, Tata McGraw Hill.

Reference Books:

1. Tanenbaum A. S. and D. Wetherall, *Computer Networks*, 5th Edition, 2013, Pearson Education.
2. William Stallings, *Data and Computer Communication,* 9th Edition, 2011, Pearson Education, Inc.

Paper V/Subject Name: Principles of Management and Organizational Behavior		**Subject Code: BSA022C505**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand about the principles of management and their application to the functioning of an organization.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** the managerial functions like planning, and have same basic knowledge on international aspect of management	**BT 2**
CO 2	**Build** the ability to direct, leadership and communicate effectively	**BT 3**
CO3	**Analyze** the behavior of individuals and groups in organizations in terms of the key factors that influence organizational behavior.	**BT 4**
CO 4	**Assess** the potential effects of organizational-level factors (such as structure, culture and change) on organizational behavior	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction To Management and Organizations**	Definition of Management – Science or Art, Manager vs. Entrepreneur, types of managers, managerial roles and skills. Evolution of Management-Scientific, human relations, system and contingency approaches, Types of Business organization, Sole proprietorship, partnership, company public and private sector enterprises, Organization culture and Environment, Current trends and issues in Management.	**18**
II	**Planning and Decision Making**	Nature and purpose of planning, planning process, types of planning, objectives, setting objectives, policies. Planning premises, Strategic Management, Planning Tools and Techniques, Decision making steps and process.	**15**
III	**Organization and Human Resource Management**	Organizing- Nature and purpose, Formal and informal organization, organization chart, organization structure, types, Line and staff authority, departmentalization, delegation of authority, centralization and decentralization, Job Design. Human Resource Management- HR Planning, Recruitment, selection, Training and Development, Performance Management, Career planning and management.	**15**
IV	**Direction and Control**	Directing-Foundations of individual and group behavior, motivation, motivation theories, motivational techniques, job satisfaction, job enrichment. Leadership- types and theories of leadership, communication, process of communication, barrier in communication, effective Communication, Communication and IT. Controlling- System and process of controlling, budgetary and non-budgetary control techniques, use of computers and IT in Management control, Productivity problems and management, control and performance, direct and preventive control, reporting.	**18**
		TOTAL	**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**

3* 22 NCH = 66 NCH	--	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Management,* Stephen P. Robbins and Mary Coulter, 13th Edition, 2017, Prentice Hall India Pvt. Ltd.
2. *Fundamentals of Management,* Stephen A. Robbins, David A. Decenzo and Mary Coulter, 9th Edition, 2016, Pearson Education India.

Reference Books:

1. Robert Kreitner and Mamata Mohapatra,*Management*, 1st Edition, 2008, Dreamtech Press.
2. Harold Koontz and Heinz Weihrich, *Essentials of Management: An International, Innovation and Leadership Perspective,* 10th Edition, 2015, Tata McGraw Hill.
3. Tripathy P. C. & Reddy P. N., *Principles of Management*, 4th Edition, 2010, Tata McGraw Hill.
4. J. P. Pathak, *Fundamentals of Management,* 1st Edition, 2014, Vikas Publishing House.
5. Robert N. Lussier, *Management Fundamentals Concepts, Applications, Skill Development,* 5th Edition, 2012, Cengage Publications.

Paper VI/Subject Name: Prompt Engineering		**Subject Code: CSE022G506**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Course Objective:

The objectives of the course are to equip students with the knowledge and practical skills required to design, optimize, and ethically apply prompt engineering techniques for Generative AI and Large Language Model based applications.

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Define** and **explain** the architecture and working principles of Generative AI and LLMs.	**BT 1 & 2**
CO 2	**Construct** effective prompts using zero-shot, few-shot, chain-of-thought, and role-based prompting methods.	**BT 3**
CO 3	**Analyse** and optimize prompts for accuracy, relevance, creativity, and safety.	**BT 4**
CO 4	**Evaluate** ethical, legal, and societal implications of AI-generated content and automation systems.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**

I	**Foundations of Generative AI and Prompt Engineering**	Introduction to Artificial Intelligence and Generative AI, Evolution of Large Language Models (LLMs), Fundamentals of Prompt Engineering, Tokens, Context Windows, Temperature, Top-p Sampling, Transformer Architecture Basics, Working of GPT, Gemini, Claude, and Open-source LLMs, Types of Prompts: Instruction Prompts, Contextual Prompts, Role-based Prompts, Conversational Prompts, Prompt Design Principles, AI-assisted Learning and Productivity Applications	**15**
II	**Advanced Prompting Techniques**	Zero-shot Prompting, Few-shot Prompting, Chain-of-Thought Prompting, Self-consistency Prompting, ReAct Framework, Tree-of-Thought Prompting, Prompt Chaining Techniques, Structured Output Generation, Prompt Templates and Reusable Frameworks, Hallucination Detection and Mitigation, AI Output Validation Techniques	**15**
III	**Multimodal AI and AI-Assisted Development**	Introduction to Multimodal AI Systems, Text-to-Image Prompting, Image Understanding and Visual Prompting, Prompting for Audio and Video Generation, AI-assisted Coding using GitHub Copilot and LLMs, Code Generation and Debugging Prompts, Prompt Engineering for Data Analysis, AI Agents and Autonomous Workflows, Prompt APIs and Integration Basics, Domain-specific Prompting: Healthcare, Education, Agriculture, Business Analytics	**15**
IV	**Ethics, Security, Evaluation, and Future Trends**	Responsible and Ethical AI, Bias and Fairness in AI Models, Privacy, Security, and Prompt Injection Attacks, AI Governance and Regulatory Frameworks, Intellectual Property and Copyright Issues, Evaluating AI-generated Outputs, Benchmarking Prompt Performance, Human-in-the-loop Systems, Emerging Trends in Prompt Engineering, Future of AI-Augmented Computing	**15**
Total			**60**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Co-intelligence: Living and working with AI,* Ethian Mollick, E., 2024, Portfolio.
2. *Demystifying artificial intelligence for the enterprise,* Geerling, W., 2023, Routledge.

Reference Books:

1. White, J., Fu, Q., Hays, S., Sandborn, M., Olea, C., Gilbert, H., Elnashar, M., Spencer-Smith, J., & Schmidt, D. C., *A prompt pattern catalog to enhance prompt engineering with ChatGPT,* 2023, Vanderbilt University.
2. Marr, B., *Generative AI in practice: 100+ amazing ways generative artificial intelligence is changing business and society,* 2023, Wiley.

6.7 Detailed Syllabus of 6th Semester

Paper I/Subject Name: Computer Networks		Subject Code: CSE022C601
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make the students understand the fundamental concepts and reference models of computer networks, apply error detection, flow control, and congestion control techniques in network protocols, etc.

Prerequisites: Concepts of Data Communication

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the layered architecture of computer networks and functions of each layer.	**BT 1 & 2**
CO 2	**Apply** subnetting, IP addressing, and routing algorithms in network design.	**BT 3**
CO 3	**Analyze** the operation of error control, flow control, and congestion control mechanisms.	**BT 4**
CO 4	**Evaluate** performance and reliability of network protocols and simulate them using tools.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Data Link Layer and Medium Access Sub-layer**	Design issues, Framing, Error detection and correction codes: checksum, CRC, hamming code, Data link protocols for noisy and noiseless channels, Sliding Window Protocols: Stop & Wait ARQ, Go-back-N ARQ, Selective repeat ARQ, Data link protocols: HDLC and PPP Static and dynamic channel allocation, Random Access: ALOHA, CSMA protocols, Controlled Access: Polling, Token Passing, IEEE 802.3 frame format, Ethernet cabling, Manchester encoding, collision detection in 802.3, Binary exponential back off algorithm	**15**
II	**Network Layer**	Design issues, IPv4 classful and classless addressing, subnetting, Routing algorithms: distance vector and link state routing, Congestion control: Principles of Congestion Control, Congestion prevention policies, Leaky bucket and token bucket algorithms	**18**
III	**Transport Layer**	Elements of transport protocols: addressing, connection establishment and release, flow control and buffering, multiplexing and de-multiplexing, crash recovery, introduction to TCP/UDP protocols and their comparison	**18**

IV	**Application Layer**	World Wide Web (WWW), Domain Name System (DNS), E-mail, File Transfer Protocol (FTP), SMTP, HTTP, Introduction to Network security	**15**
		TOTAL	**66**

Computer Networks Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 20 Laboratory experiments based on the following-

1. To study various topologies for establishing computer networks.
2. To learn the usage of various basic tools (crimping, krone etc.) used in establishing a LAN.
3. To familiarize with switch and hub used in networks.
4. To learn the usage of connecters and cables (cabling standards) used in networks
5. To make certain copper and fiber patch cords using different standards.
6. To familiarize with routers & bridges
7. Use commands like ping, ipconfig for trouble shooting network related problems.
8. NIC Installation & Configuration (Windows/Linux)
9. TCP/UDP Socket Programming
10. Multicast & Broadcast Sockets
11. Develop a program to compute the Hamming Distance between any two code words.
12. Develop a program to compute checksum for an ‗m' bit frame using a generator polynomial.
13. IPC (Message queue)
14. Implementation of a Prototype Multithreaded Server
    a. Implementation of o Data Link Layer Flow Control Mechanism (Stop & Wait, Sliding Window)
    b. Data Link Layer Error Detection Mechanism (Cyclic Redundancy Check)
    c. Data Link Layer Error Control Mechanism (Selective Repeat, Go Back N)

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Data and Computer Communication*, William Stallings, 10th Edition, 2013, PHI.
2. *Data Communications and Networking*, Behrouz A Forouzan, 4th Edition, 2017, Tata McGraw Hill
3. *Computer Networks*, Tannenbaum, 3rd Edition, 1996, Pearson Education.

Reference Books:

1. L.L. Peterson & B.S. Davie, *Computer Networks: A Systems Approach*, 5th Edition, 2011, Morgan Kaufmann
2. Anuranjan Misra, *Computer Networks*, 2006, Acme Learning, Morgan Kaufman Publication, New Delhi

Paper II/Subject Name: Compile Design		Subject Code: CSE022C602
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective

The objectives of the course are to provide the basic concept of compiler design and make the students understand concepts of lexical analyser, code generation and code optimization techniques.

Prerequisites: Concepts of Formal Language and Automata Theory

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** the structure and phases of a compiler and the role of each phase.	**BT 1 & 2**
CO 2	**Construct** finite automata and regular expressions for lexical analysis.	**BT 3**
CO 3	**Analyze** various parsing techniques like LL, LR, SLR, and LALR parsing.	**BT 4**
CO 4	**Assess** intermediate code and code optimization techniques.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Overview of the Translation Process, A Simple Compiler, Difference between interpreter, assembler and compiler. Overview and use of linker and loader, types of Compiler, Analysis of the Source Program, The Phases of a Compiler, Cousins of the Compiler, The Grouping of Phases, Lexical Analysis, Hard Coding and Automatic Generation Lexical Analyzers, Front-end and Back-end of compiler, pass structure	**15**
II	**Lexical Analyzer and Parsing Theory**	Introduction to Lexical Analyzer, Input Buffering, Specification of Tokens, Recognition of Tokens, A Language for Specifying Lexical Analyzers, Finite Automata from a Regular Expression, Design of a Lexical Analyzer Generator, Optimization of DFA, Top Down and Bottom-up Parsing Algorithms, Top-Down Parsing, Bottom-Up Parsing, Operator-Precedence Parsing, LR Parsers, Using Ambiguous Grammars, Parser Generators, Automatic Generation of Parsers. Syntax-Directed Definitions, Construction of Syntax Trees, Bottom-Up Evaluation of S-Attributed Definitions, L-Attributed Definitions, and syntax directed definitions and translation schemes.	**18**
III	**Intermediate Code Generation and Run-Time**	Different Intermediate Forms, Syntax Directed Translation Mechanisms and Attributed Mechanisms and Attributed Definition, Source Language Issues, Storage Organization,	**18**

	Memory Management	Storage-Allocation Strategies, and Access to Non local Names, Parameter Passing, Symbol Tables, and Language Facilities for Dynamic Storage Allocation, Dynamic Storage Allocation Techniques. Error Detection & Recovery, Ad-Hoc and Systematic Methods	
IV	**Code Optimization and Code Generation**	Global Data Flow Analysis, A Few Selected Optimizations like Command Sub Expression Removal, Loop Invariant Code Motion, Strength Reduction etc. Issues in the Design of a Code Generator, The Target Machine, Run-Time Storage Management, Basic Blocks and Flow Graphs, Next-Use Information, A Simple Code Generator, Register Allocation and Assignment, The DAG Representation of Basic Blocks, Peephole Optimization, Generating Code from DAGs, Dynamic Programming Code-Generation Algorithm, Code Generators.	**15**
TOTAL			**66**

Compiler Design Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Write a Lex program to recognize valid identifiers, keywords, and numbers.
2. Implement a Lex program to compute the number of lines, words, and characters in a file.
3. Write a Yacc program to check the syntax of simple arithmetic expressions.
4. Develop a Yacc parser to recognize nested if-else and while constructs.
5. Construct a parse tree for a given arithmetic expression.
6. Generate three-address code (TAC) for assignment and conditional statements.
7. Implement backpatching for Boolean expressions and flow-of-control statements.
8. Write code to perform constant folding and dead code elimination on a given intermediate code.
9. Implement a symbol table with scope handling.
10. Mini-project: Design a mini-compiler for arithmetic and control expressions.

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Compilers: Principles, Techniques, and Tools,* A.V. Aho, Monica Lam, Ravi Sethi, and J.D. Ullman, 2nd Edition, 2006, Addison Welsley
2. *Engineering a Compiler,* K.D. Cooper, and Linda Torczon, 2nd Edition, 2011, Morgan Kaufmann

Reference Books:

1. K.C. Louden, *Compiler Construction: Principles and Practice,* 1st Edition, 1997, Cengage Learning

2. D. Brown, J. Levine, and T. Mason, *LEX and YACC,* 2nd Edition, 2011, O"Reilly Media

Paper III/Subject Name: Software Engineering		**Subject Code: CSE022C603**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain the fundaments of software engineering principles and practices, including project management, configurations management, requirements definition, system analysis, design, testing, and deployment.

Prerequisites: None

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **demonstrate** the fundamental concepts of software engineering, software process models, and development life cycles.	**BT 1 & 2**
CO 2	**Identify** software requirements and develop software specifications using functional and non-functional requirements.	**BT 3**
CO 3	**Analyze** risk factors, cost estimation, and scheduling using various estimation and modeling techniques.	**BT 4**
CO 4	**Evaluate** and select appropriate software testing strategies, maintenance models, and quality assurance standards.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Hours**
I	**Introduction to Process Models and Software Requirement Specification**	Importance of Software Project Management, Activities Methodologies, Categorization of Software Projects, Setting objectives, Software life cycle models: Waterfall, prototyping, Evolutionary, Spiral models and Agile Model. Software Requirements: Functional and non-functional requirements, user requirements, system requirements, interface specification, the software requirements document. Requirements engineering process: Feasibility studies, requirements elicitation and analysis, requirements validation, requirements management.	**18**
II	**Software Process Management, Activity Planning and Agile Development**	Project planning and control, Effort and Cost estimation techniques-LOC, Function Point, COCOMO, project scheduling using PERT and GANTT charts, Critical path (CRM) method, cost-time relations: Rayleigh-Norden results, Staffing Pattern,S oftware configuration management, Introduction to Agility- Agile methods – Extreme Programming – SCRUM – Managing interactive processes.	**15**
III	**Software Design and Risk Estimation**	Basics of Software Design, Procedural Design Methodology, Modularity, Cohesion, Coupling, DFD and Structure Chart, Object-Oriented concepts, Introduction to UML: Class and	**15**

		interaction Diagrams, Object-Oriented Analysis and Design, Object-oriented Software Modelling. Risk Management-Risk Identification, Risk Assessment, Risk Containment	
IV	**Software Testing, Maintenance and Reuse**	Software testing fundamentals-Internal and external views of Testing-white box testing – basis path testing-control structure testing-black box testing-Regression Testing – Unit Testing – Integration Testing – Validation Testing – System Testing And Debugging –Software Implementation Techniques: Coding practices-Refactoring-Maintenance and Reengineering-BPR model-Reengineering process model-Reverse and Forward Engineering. Characteristics of Software Maintenance, Software Reverse Engineering, Software Maintenance Process Models, Estimation of maintenance cost, Software Reuse.	**18**
Total			**66**

Credit Distribution		
Lecture/ Tutorial	**Practicum/ Tutorial**	**Experiential Learning**
3* 22 NCH = 66 NCH	-	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Software Project Management,* Bob Hughes, Mike Cotterell and Rajib Mall, 5th Edition, 2012, Tata McGraw Hill, New Delhi

Reference Books:

1. Kieron Conway, *Software Project Management: From Concept to Deployment,* 1st Edition, 2000, Dreamtech Press.
2. S. A. Kelkar, *Software Project Management: A Concise Study,* 3rd Edition, 2012, PHI Publication.

Paper IV/Subject Name: Mobile Computing		**Subject Code: CSE022D601**
Paper Type: PEC-I		**Specialization: Network Engineering**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to teach the concept of mobile computing paradigm, its applications and limitations, typical mobile networking Infrastructure through a popular GSM protocol.

Prerequisites: Concepts of Computer Networks

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **demonstrate** mobile technologies in terms of hardware, software, and communications and describe how mobile technology functions to enable other computing technologies.	**BT 1 & 2**
CO 2	**Utilize** mobile computing nomenclature to describe and analyze existing mobile computing frameworks and architectures.	**BT 3**
CO 3	**Analyze** any new technical issues related to new paradigm and come up with a solution(s).	**BT 4**
CO 4	**Evaluate** the effectiveness of different mobile computing frameworks.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Mobile Communications, Mobile Computing – Paradigm. Promises/Novel Applications and impediments and Architecture; Mobile and Hand held Devices, LirMatioris of Mobile and Handhold Devices. GSM — Services. System Architecture. Radio Interlaces, Protocols. Localization, Calling, Handover, Security, New Data Services, GPRS, CSHSD, DECT.	**22**
II	**Wireless Medium Access Control (MAC)**	Motivation for a specialized MAC (Hidden and exposed terminals. Near and far terminals), SOMA, FDMA TOMA, COMA, Wireless LAN/(IEEE802.11), Mobile Network Layer IP end Mobile IP Network Layers, PacketDelivery and Handover Management, Location Management, Registration. Tunneling and Encapsulation, Route Optimization, DHCR Mobile Transport Layer: Conventional TCP/IP Protocols, Indirect TCP, Snooping TCP, Mobile TCP, Other Transport Layer Protocols for Mobile Networks. Database Issues: Database Hoarding & Caching Techniques. Client-Server Computing a Adaptation, Transnational Models, Query processing, Data Recovery Process & QoS Issues	**22**
III	**Data Dissemination and Synchronization**	Communications Asymmetry. Classification of Data Delivery Mechanisms. Data Dissemination, Broadcast Models. Selective Tuning and Indexing Methods. Data Synchronization – Introduction. Software and Protocols.	**22**
IV	**Mobile Ad hoc Networks (MANETs)**	Introduction, Applications a Challenges of a MANET Routing, Classification of Routing Algorithms. Algorithms such as DSR. AODV. DSDV. etc. , Mobile Agents. Service Discovery. Protocols and Platforms for Mobile Computing : WAP, Bluetooth. XML. J2ME. JavaCard. PaImOS, Windows CE. SymbianOS. Unix for Mobile Devices. Android.	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Mobile Communication*, Joctien Schuller, 2nd Edition, 2009, Addison-Wesley
2. *MobiIe Computing*, Raj Kamal, 2nd Edition, 2007,Oxford University Press.

Reference Books:

1. Jochen Sdiiler,*Mobile Communications*, 2nd Edition, 2004,Addison-Wesley.
2. Stomenovlc and Cacute, Handbook of Wireless Networks and Mobile Computing, 2002, Wiley.
3. Reza Behravanfar, *Mobile Computing Principles: Designing and Developing Mobile Applications with UML and XML,* 2004, Cambridge University Press.

Paper V/Subject Name: Big Data Analytics		**Subject Code: CSE022D602**
Paper Type: PEC-II		**Specialization: Network Engineering**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce students to the concepts, tools, and frameworks of Big Data, focusing on storage, processing, analytics of large-scale datasets and hands-on skills using technologies like Hadoop and Spark for real-world data-driven problem solving.

Prerequisites: Concepts of Databases and Networks

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **understand** the fundamentals, challenges, and technologies of big data.	**BT 1 & 2**
CO 2	**Apply** Hadoop ecosystem tools for distributed data storage and processing.	**BT 3**
CO 3	**Analyze** large-scale data using MapReduce and Apache Spark frameworks.	**BT 4**
CO 4	**Evaluate** various big data analytics techniques for structured and unstructured data.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction to Big Data and Hadoop Framework**	Big Data – Definition, Characteristic Features – Big Data Applications - Big Data vs Traditional Data - Risks of Big Data - Structure of Big Data - Challenges of Conventional Systems - Web Data – Evolution of Analytic Scalability - Evolution of Analytic Processes, Tools and methods - Analysis Vs Reporting - Modern Data Analytic Tools. Distributed File Systems - Large-Scale File System Organization – HDFS concepts - MapReduce Execution, Algorithms using MapReduce, Matrix-Vector Multiplication – Hadoop YARN	**22**
II	**Data Analysis**	Statistical Methods: Regression modelling, Multivariate Analysis - Classification: SVM & Kernel Methods - Rule Mining - Cluster Analysis, Types of Data in Cluster Analysis, Partitioning Methods, Hierarchical Methods, Density Based Methods, Grid Based Methods, Model Based Clustering Methods, Clustering High Dimensional Data - Predictive Analytics – Data analysis using R	**22**
III	**Mining Data Streams**	Streams: Concepts – Stream Data Model and Architecture - Sampling data in a stream - Mining Data Streams and Mining Time-series data - Real Time Analytics Platform (RTAP) Applications - Case Studies - Real Time Sentiment Analysis, Stock Market Predictions.	**22**
IV	**Big Data Frameworks**	Introduction to NoSQL – Aggregate Data Models – Hbase: Data Model and Implementations – Hbase Clients – Examples – Cassandra: Data Model – Examples – Cassandra Clients – Hadoop Integration. Pig – Grunt – Pig Data Model – Pig Latin – developing and testing Pig Latin scripts. Hive – Data Types and File Formats – HiveQL Data Definition – HiveQL Data Manipulation – HiveQL Queries	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Taming the Big Data Tidal Wave: Finding Opportunities in Huge Data Streams with Advanced Analytics,* Bill Franks, 1st Edition, 2012, Wiley and SAS Business Series.
2. David Loshin, "Big Data Analytics: From Strategic Planning to Enterprise Integration with Tools, Techniques, NoSQL, and Graph", 2013

Reference Books:

1. Michael Berthold, David J. Hand, *Intelligent Data Analysis,* 2nd Edition, 2007, Springer.
2. Michael Minelli, Michelle Chambers, and Ambiga Dhiraj, *Big Data, Big Analytics: Emerging Business Intelligence and Analytic Trends for Today's Businesses*, 1st Edition, 2013, Wiley.
3. P. J. Sadalage and M. Fowler, *NoSQL Distilled: A Brief Guide to the Emerging World of Polyglot Persistence*", 1st Edition, 2012, Addison-Wesley Professional.
4. Richard Cotton, *Learning R- A Step-by-step Function Guide to Data Analysis*, 1st Edition, 2013, O'Reilly Media.

Paper IV/Subject Name: Introduction to Semiconductor Devices		**Subject Code: CSE022D603**
Paper Type: PEC-I		**Specialization: Semiconductors**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the core physical principles and behavior of semiconductor materials and their importance in electronic devices and apply the working of fundamental semiconductor devices such as diodes, BJTs, and MOSFETs in electronic circuits.

Prerequisites: Concepts of Basic Physics, Digital Logic and Design

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the properties and behavior of semiconductor materials.	**BT 1 & 2**
CO 2	**Illustrate** the functioning and applications of Bipolar Junction Transistors (BJTs).	**BT 3**
CO 3	**Analyze** the operation of PN junction diodes and Zener diodes, FETs and MOSFETs.	**BT 4**
CO 4	**Evaluate** circuit problems involving semiconductor devices in practical applications.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Semiconductor Fundamentals**	Crystal structure of solids: unit cells, lattice structures, crystal defects, Energy bands in solids: conductors, semiconductors, insulators, Types of semiconductors: intrinsic and extrinsic (n-type, p-type), Carrier concentration in semiconductors: Mass-action law, Fermi-Dirac statistics, Fermi level positioning for intrinsic and extrinsic semiconductors, Carrier transport mechanisms: Drift current and mobility, Diffusion current and diffusion coefficient, Generation and recombination of charge carriers, Continuity equation, Einstein relation and quasi-Fermi levels	**22**
II	**PN junction Diodes**	PN junction formation and equilibrium, Biasing of PN junction: forward and reverse bias, V-I characteristics and diode equation, Transition and diffusion capacitance, Zener diode: breakdown	**22**

		mechanism, characteristics, voltage regulation, Diode models: ideal, practical with resistances, Diode circuits: Half-wave and full-wave rectifiers (with and without filters): Bridge rectifier, Clippers and clampers, Peak detectors and voltage doublers, Light Emitting Diodes (LED), Photodiodes, Solar Cells	
III	**Bipolar Junction Transistors**	BJT structure and operation (NPN and PNP), Biasing techniques: fixed bias, collector-to-base bias, voltage divider bias, Input/output characteristics of CB, CE, CC configurations, DC load line and Q-point analysis, Thermal runaway and stability factor, Small-signal model: h-parameters, BJT as an amplifier: voltage gain, input/output resistance, BJT as a switch: saturation and cutoff operation, Applications in digital and analog switching	**22**
IV	**Field Effect Transistors and MOSFETs**	JFET: structure, working principle, characteristics, Shockley's equation and parameters (IDSS, VP), Biasing of JFETs: self-bias, voltage divider bias, MOSFET: Enhancement and Depletion modes, N-channel and P-channel MOSFETs, I-V characteristics and threshold voltage, Comparison of BJT and FET, FET/MOSFET small-signal models, CMOS basics and logic gates, Applications in logic circuits, amplifiers, and analog switches	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Electronic Devices And Circuits*, S Salivahanan, N Suresh Kumar, 5th Edition, 2022, McGraw Hill Education
2. *Electronic Devices and Circuit Theory,* R.L. Boylestad & Louis Nashelsky, 1st Edition, 2012, Pearson Education.

Reference Books:

1. Donald A. Neamen, *Semiconductor Physics and Devices: Basic Principles,* 4th Edition, 2011, McGraw Hill Education
2. Sedra & Smith, *Microelectronic Circuits,* 6th Edition, 2009, Oxford University Press

Paper V/Subject Name: VLSI Design		**Subject Code: CSE022D604**
Paper Type: PEC-II		**Specialization: Semiconductors**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamentals of CMOS technology and digital VLSI design flow.

Prerequisites: Basics of Digital Logic, Computer Organization

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **understand** the operation and characteristics of MOSFETs and CMOS logic gates.	**BT 1& 2**
CO 2	**Apply** layout design rules to develop physical designs for digital circuits.	**BT 3**
CO 3	**Analyze** and design CMOS-based combinational and sequential logic circuits.	**BT 4**
CO 4	**Evaluate** timing, power, and area trade-offs in digital VLSI systems.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**MOS devices and CMOS Technology Fundamentals**	Basics of VLSI systems and design flow, MOS transistor structure, operation and characteristics, Threshold voltage, body effect, channel length modulation, NMOS vs PMOS, CMOS inverter design, DC transfer characteristics and noise margins, Introduction to scaling and Moore's Law, CMOS fabrication process (p-well, n-well, twin-tub, SOI), Stick diagrams and layout design rules (λ-based)	**22**
II	**CMOS Logic Design**	Static CMOS logic: NAND, NOR, XOR, complex gates, Transmission gate logic, Dynamic CMOS logic: Domino and NORA logic, Logic design styles: complementary, pass transistor, ratioed logic, Power dissipation in CMOS: static and dynamic, Delay estimation and logical effort, Fan-in, fan-out, and sizing of transistors, Subsystem design: adders, multiplexers, decoders	**22**
III	**Sequential Logic and Timing Analysis**	Latches and Flip-flops (D, T, SR, JK), Static and dynamic sequential circuits, Clocking strategies, clock skew, setup and hold time, Sequential logic timing, pipelining, and retiming, FSM Design using VLSI methodology, Memory design: SRAM, DRAM basics, Timing models, delay calculations, and critical path analysis	**22**
IV	**HDL Modeling and VLSI Tools**	Introduction to HDL: Verilog / VHDL syntax and structure, Modeling combinational and sequential circuits, Behavioral, structural, and dataflow modelling, Testbench development and simulation, Design synthesis and technology mapping, Floorplanning and placement, Introduction to EDA tools (Cadence/Synopsys/Xilinx/ModelSim), Case Study: Design and simulation of a simple digital system	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Introduction to VLSI Circuits and Systems,* John .P. Uyemura, 1st Edition, 2001, John Wiley
2. *Basic VLSI Design*, Douglas A Pucknell, Kamran Eshraghian, 3rd Edition, 1995, Prentice Hall Indi

Reference Books:

1. Wayne Wolf, *Modern VLSI Design*, 3rd Edition, 1997, Pearson Education

Paper IV/Subject Name: Social Network Analysis		**Subject Code: CSE022D605**
Paper Type: PEC-I		**Specialization: General**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamental concepts of graph theory and their applications in analysing social structures, apply algorithms for social network mining and link prediction, etc.

Prerequisites: Basic knowledge of Graph Theory, Probability and Statistics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the fundamentals of social network structures and graph properties. Understand	**BT 1 & 2**
CO 2	**Apply** algorithms for link prediction and influence propagation.	**BT 3**
CO 3	**Analyze** real-world networks using graph metrics like centrality and clustering.	**BT 4**
CO 4	**Evaluate** network models and community detection algorithms.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Semiconductor Fundamentals**	Overview of social networks and their types, Graph preliminaries: nodes, edges, degrees, walks, paths, connectivity, Types of graphs: directed, undirected, weighted, bipartite, multigraph, Graph representations: adjacency matrix, edge list, adjacency list, Real-world network examples: Facebook, Twitter, LinkedIn, citation	**22**

		networks, Network data collection and visualization tools (Gephi, NetworkX)	
II	**Structural Properties and Network Metrics**	Degree distribution, average path length, diameter, Clustering coefficient and transitivity, Centrality measures: Degree centrality, Betweenness centrality, Closeness centrality, Eigenvector centrality, Assortativity, reciprocity, density, and connectivity, Triadic closure and small-world phenomenon	**22**
III	**Community Detection and Network Models**	Community structures in graphs, Modularity and methods for community detection: Girvan–Newman algorithm, Label Propagation, Louvain method, Graph partitioning and hierarchical clustering, Network evolution and growth models: Erdős–Rényi Random Graph, Watts–Strogatz Small-World Model, Barabási–Albert Preferential Attachment Model, Structural holes and bridges	**22**
IV	**Algorithms and Applications**	Link prediction techniques (common neighbors, Jaccard, Adamic-Adar, Katz), Influence maximization and information diffusion models: Independent Cascade Model, Linear Threshold Model, Homophily, propagation of innovation, Recommendation systems using networks, Applications in epidemiology, marketing, cyber security, and fake news detection, Case study: Twitter influence graph, Facebook friend graph analysis	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Networks, Crowds, and Markets: Reasoning About a Highly Connected World,* David Easley & Jon Kleinberg, 2010, Cambridge University Press
2. *Social Network Analysis: Methods and Applications,* S. Wasserman & K. Faust, 1994, Cambridge University Press
3. *Social Network Data Analytics,* Charu C. Aggarwal, 11th Edition, 2011, Springer

Reference Books:

1. Mark Newman, *Networks: An Introduction,* 1st Edition, 2010, Oxford University Press
2. Robert A. Hanneman, *Introduction to Social Network Methods,* 2005, University of California

Paper V/Subject Name: Digital Image Processing		**Subject Code: CSE022D606**
Paper Type: PEC-II		**Specialization: General**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide the students with an overall insight to the process of Digital Image Processing and teach various image enhancement techniques and image representation process.

Prerequisites: Basic Concepts of Programming and Mathematics

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and i**llustrate** the importance of image transforms, different types of image transforms and their properties, image compression, etc.	**BT 1 & 2**
CO 2	**Experiment with** the different image enhancement techniques	**BT 3**
CO 3	**Analyze** the different causes for image degradation.	**BT 4**
CO 4	**Assess** the various techniques of image segmentation	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Digital Image fundamentals**	Introduction – Origin – Steps In Digital Image Processing – Components – Elements Of Visual Perception – Image Sensing And Acquisition – Image Sampling And Quantization – Relationships Between Pixels – Color Models.	**22**
II	**Image Enhancement**	Spatial Domain: Gray Level Transformations – Histogram Processing – Basics Of Spatial Filtering–Smoothing And Sharpening Spatial Filtering – Frequency Domain: Introduction To Fourier Transform – Smoothing And Sharpening Frequency Domain Filters – Ideal, Butterworth And Gaussian Filters. Image restoration and segmentation: Noise Models – Mean Filters – Order Statistics – Adaptive Filters – Band Reject Filters – Band Pass Filters – Notch Filters – Optimum Notch Filtering – Inverse Filtering – Wiener Filtering Segmentation: Detection Of Discontinuities–Edge Linking And Boundary Detection – Region Based Segmentation- Morphological Processing- Erosion And Dilation.	**22**
III	**Wavelets and Image Compression**	Wavelets – Subband Coding – Multiresolution Expansions – Compression: Fundamentals – Image Compression Models – Error Free Compression – Variable Length Coding – Bit-Plane Coding – Lossless Predictive Coding – Lossy Compression – Lossy Predictive Coding – Compression Standards.	**22**
IV	**Image representation and recognition**	Boundary Representation – Chain Code – Polygonal Approximation, Signature, Boundary Segments – Boundary Description – Shape Number – Fourier Descriptor, Moments- Regional Descriptors –Topological Feature, Texture – Patterns And Pattern Classes – Recognition Based On Matching	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Digital Image Processing*, Rafael C. Gonzales, Richard E. Woods, 3rdEdition, 2010, Pearson Education, 2010.

Reference Books:

1. Rafael C. Gonzalez, Richard E. Woods, Steven L. Eddins, *Digital Image Processing Using MATLAB,*3rdEdition, 2011, Tata Mc Graw Hill Pvt. Ltd.
2. Anil Jain K, *Fundamentals Of Digital Image Processing*,1st Edition, 1988, PHI Learning Pvt. Ltd.
3. Willliam K Pratt, *Digital Image Processing*, 4th Edition, 2002, John Willey
4. Malay K. Pakhira, *Digital Image Processing and Pattern Recognition*, 1stEdition, 2011, PHI Learning Pvt. Ltd.

Paper VI/Subject Name: Entrepreneurship and Startups		**Subject Code: CSE022G606**
L-T-P-C – 3-0-0-3	**Credit Units: 03**	**Scheme of Evaluation: T**

Course Objective:

The objectives of the course are to develop entrepreneurial mindset, innovation capabilities, and startup management skills among students. The course aims to familiarize learners with the startup ecosystem, business model development, funding mechanisms, intellectual property rights, and technology commercialization

Prerequisites: None

Course Outcomes

On successful completion of the course, the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** entrepreneurship, innovation, and startup ecosystems.	**BT 1 & 2**
CO 2	**Identify** business opportunities and develop sustainable business models.	**BT 3**
CO 3	**Analyze** startup funding, legal, and intellectual property aspects.	**BT 4**
CO 4	**Develop** and present a startup business plan for technology-driven ventures.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Entrepreneurship and Startup Ecosystem**	Concept, characteristics, and importance of entrepreneurship, Entrepreneurial mindset and competencies, Types of entrepreneurships: Technology, Social, Rural, and Digital Entrepreneurship, Innovation and Design Thinking fundamentals,	**12**

		Startup ecosystem in India, Government initiatives: Startup India, Make in India, Digital India, Atal Innovation Mission, Success and failure stories of startups	
II	**Opportunity Identification and Business Model Development**	Creativity and idea generation techniques, Problem identification and opportunity recognition, Market research and customer discovery, Value proposition design, Business Model Canvas (BMC) Lean Startup methodology, Minimum Viable Product (MVP), Customer validation and product-market fit	**12**
III	**Startup Formation, Funding, and Intellectual Property**	Steps in starting a venture, Forms of business organization, Startup registration and legal compliance, Intellectual Property Rights (IPR): Patents, Copyrights, Trademarks, Trade Secrets, Technology transfer and commercialization, Sources of startup funding: Bootstrapping, Angel Investors, Venture Capital, Crowdfunding, Government grants and incubators, Financial planning and revenue models	**11**
IV	**Startup Growth, Scaling, and Business Planning**	Marketing strategies for startups, Digital marketing and social media branding, Team building and leadership in startups, Risk management and business ethics, Scaling strategies for technology startups, Startup valuation basics, Pitch deck preparation, Business plan development and investor pitching	**10**
Total			**45**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3*20 NCH = 60 NCH	-	30 NCH (Problem Solving, Internship, Seminar, Case Study, Discussion)

Text Book:

1. *Entrepreneurship*, Robert D. Hisrich, Michael P. Peters and Dean A. Shepherd, 9th Edition, 2012, McGraw-Hill.
2. *Zero to One: Notes on Start Ups, or How to Build the Future,* Peter Thiel, 1st Edition, 2014, Crown Business.

Reference Books:

1. Eric Ries, *The Lean Startup*, 1st Edition, 2011, Crown Publishing.

6.8 Detailed Syllabus of 7th Semester

Paper I/Subject Name: Introduction to Artificial Intelligence		Subject Code: CSE022C701
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students learn the basic concepts of Artificial Intelligence along with its problem-solving techniques.

Prerequisites: Concepts of Mathematics, Programming Languages, Data Analytic Techniques

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Interpret** and manipulate a given problem in the language/framework of different AI methods.	**BT 2**
CO 2	**Identify** problems that are amenable to solution by AI methods.	**BT 3**
CO 3	**Analyze** basic AI algorithms for their applications.	**BT 4**
CO 4	**Evaluate** the various algorithms as per their applications	**BT 5**

Detailed Syllabus

Modules	Topics	Course Contents	Hours
I	**Introduction and Production Systems**	Introduction to AI-Problem formulation, Problem Definition - Production systems, Control strategies, Search strategies. Problem characteristics, Production system characteristics - Specialized production system- Problem solving methods - Problem graphs, Matching, Indexing and Heuristic functions - Hill Climbing-Depth first and Breath first, Constraints satisfaction - Related algorithms, Measure of performance and analysis of search algorithms.	**22**
II	**Knowledge Representation and Inference**	Game playing - Knowledge representation, Knowledge representation using Predicate logic, Introduction to predicate calculus, Resolution, Use of predicate calculus, Knowledge representation using other logic-Structured representation of knowledge Knowledge representation -Production based system, Frame based system. Inference - Backward chaining, Forward chaining, Rule value approach, Fuzzy reasoning - Certainty factors, Bayesian Theory-Bayesian Network-Dempster - Shafer theory.	**22**
III	**Planning and Machine Learning**	Basic plan generation systems - Strips -Advanced plan generation systems – K strips -Strategic explanations -Why,	**22**

		Why not and how explanations. Learning- Machine learning, adaptive Learning	
IV	**Expert Systems**	Expert systems - Architecture of expert systems, Roles of expert systems - Knowledge Acquisition – Meta knowledge, Heuristics. Typical expert systems - MYCIN, DART, XOON, Expert systems shells.	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *AI: A Modern Approach,* Stuart Russel and Peter Norvig, 2nd Edition, 2007, Pearson Education
2. *Artificial Intelligence,* Kevin Night, Elaine Rich, Nair B., 3rd Edition, 2008, Mc Graw Hill
3. *Introduction to AI and ES*, Dan W. Patterson, 3rd Edition, 2007, Pearson Education.

Reference Books:

1. Peter Jackson, *Introduction to Expert Systems*, 3rd Edition, 2007, Pearson Education
2. Deepak Khemani, *Artificial Intelligence*, 2013, Tata Mc Graw Hill Education

Paper II/Subject Name: Web Technology		**Subject Code: CSE022C702**
L-T-P-C – 3-0-2-4	**Credit Units: 04**	**Scheme of Evaluation: TP**

Objective:

The objectives of the course are to provide knowledge on the basic web concepts, scripting languages and Internet protocols.

Prerequisites: Concepts of Databases and Object-Oriented Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **understand** the basic concept of web development	**BT 1 & 2**
CO 2	**Apply** the concepts learnt to develop simple web applications	**BT 3**
CO 3	**Assess** two web applications based on various design factors.	**BT 4**
CO 4	**Evaluate** the working of various web applications.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction, To Web Technology**	World Wide Web: Introduction to TCP/IP and WAP, DNS, Email, TelNet, HTTP and FTP. Introduction to Browser and search engines, Working of the search engines, Miscellaneous Web Browser details, Introduction to Web Servers: Features of web servers, caching, case study-IIS, Apache, Configuring web servers.Internet Principles – Basic Web Concepts – Client/Server model – retrieving data from Internet – HTM and Scripting Languages – Standard Generalized Mark –up languages – Next Generation – Internet –Protocols and Applications.	**18**
II	**HTML,CSS, Java Script**	Web Pages - types and issues, tiers; comparisons of Microsoft and java technologies, WWW-Basic concepts, web client and web server, http protocol (frame format), universal resource locator (URL), HTML different Tags, sections, image & pictures, listings, tables, frame, frameset, form. The need of dynamic web pages; an overview of DHTML, cascading style sheet (CSS), comparative studies of different technologies of dynamic page creation.Java Script : Data types, variables, operators, conditional statements, array object, date object, string object, Dynamic Positioning and front end validation, creating rollovers, building smarter forms, Event Handling, working with cookies, DOM, node and objects, creating sliding menu, pop-up menu, slideshow with caption	**18**
III	**XML and AJAX**	XML – Server side includes – communication – DTD – Vocabularies – DOMmethods – Introduction of XML, Validation of XML documents, DTD, Ways to use XML, XML for data files, HTML Vs XML, Embedding XML into HTML documents, Converting XML to HTML for Display, Rewriting HTML as XML, Firewalls– Proxy Servers.AJAX technologies, Action, XML HttpRequest database operations, security, issues	**15**
IV	**J2SE, J2EE, Servlet and JSP**	Data Types, Arrays, Type Casting, Classes and Objects, Inheritance, Interfaces, Exception Handling, Multithreading,J2EE as a framework, Client Server Traditional model, Comparison amongst 2-tier, 3-tier and N-tier Architectures, Thin and Thick Clients. J2EE Servlet 2.x Specification, Writing small Servlet Programs, Deployment Descriptor, Inter Servlet Collaboration, Session: Definition, State on web, Different ways to track sessions.JSP Technology Introduction-JSP and Servlets- Running JSP Applications Basic JSP- JavaBeans Classes - Support for the Model-View- Controller Paradigm- Case Study- Related Technologies.	**15**
TOTAL			**66**

Web Technology Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Web page design: Designing web pages with HTML- use of tags, hyperlinks, URLs, tables, text formatting, graphics & multimedia, imagemap, frames and forms in web pages.
2. Use of Cascading Style Sheet in web pages.
3. Creating interactive and dynamic web pages with JavaScript: JavaScript overview; constants, variables, operators, expressions & statements; user-defined & built-in functions; client-side form validation; using properties and methods of built-in objects.
4. Extensible Markup Language (XML): Introduction- using user-defined tags in web pages; displaying XML contents; XML DTDs; use of XSL.
5. Server-side scripting: overview of CGI, ASP, and JSP.
6. Server side scripting using PHP; PHP basics, HTML form data handling, Web database connectivity- introduction to ODBC; PHP with database connectivity.
   Exposure to Advanced Web Technologies (as far as possible; not to be made compulsory): Distributed Object based models- DCOM, CORBA, EJB; Web services and Related Technologies- ISAPI, SOAP, UDDI, WSDL; Other Advanced Web Technologies- AJAX, ISAPI, .NET. Web Security

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Internet and World Wide Web How to program*,Deitel H.M. and Deitel P.J, 4th Edition, 2012, Pearson International, New Delhi.
2. *Web Technology,* Gopalan N.P. and Akilandeswari J., 2nd Edition, 2014. Prentice Hall of India.
3. *Java How to Program*, Paul Dietel and Harvey Deitel, 8th Edition, 2014, Prentice Hall of India.

Reference Books:

1. Uttam K. Roy, *Web Technologies*, 2011, Oxford University Press.
2. Godbole A. S. & Kahate A., TMH, *Web Technologies*, 2ndEdition, 2006, TMH

Paper III/Subject Name: Cloud Computing		**Subject Code: CSE022D701**
Paper Type: PEC-III		**Specialization: Network Engineering**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the fundamental concepts, architecture, and service models of cloud computing, explore virtualization and its role in enabling cloud scalability, etc.

Prerequisites: Concepts of Networks and Operating Systems

Course Outcomes

On successful completion of the course the students will be able to:		
Sl No	**Course Outcome**	**Blooms Taxonomy Level**

CO 1	**Describe** and **explain** the core concepts and architecture of cloud computing.	**BT 1 & 2**
CO 2	**Illustrate** the use of virtualization and resource provisioning.	**BT 3**
CO 3	**Analyze** cloud deployment models and services (IaaS, PaaS, SaaS).	**BT 4**
CO 4	**Evaluate** cloud security and performance trade-offs.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Overview of Cloud Computing, History, Evolution, and Characteristics, On-demand computing and utility models, Cloud computing architecture: Client, Cloud Service, Cloud Infrastructure, Cloud deployment models: Public, Private, Hybrid, Community, Cloud service models: IaaS, PaaS, SaaS, FaaS, Cloud reference architecture (NIST model), Benefits and limitations of cloud computing	**22**
II	**Virtualization and Resource Management**	Introduction to virtualization: types and uses, Hypervisors: Bare-metal (Type 1), Hosted (Type 2), Virtual Machines (VMs) vs Containers (Docker), Resource pooling and elasticity, Virtual memory and virtual storage concepts, VM migration and load balancing, Cloud storage: architecture, models (object, block, file), Case study: Virtualization in AWS EC2, Azure VMs	**22**
III	**Cloud Services, Programming and Management**	Service orchestration and provisioning, Cloud APIs and web services (REST, SOAP), Programming frameworks: Hadoop, MapReduce, Spark on Cloud, Serverless computing (AWS Lambda, Google Cloud Functions), Cloud application development (Python, Node.js, etc.), Auto-scaling and elasticity, Cloud pricing models: pay-as-you-go, reserved instances, Hands-on: Deploying apps on AWS / GCP / Azure	**22**
IV	**Security, Privacy and Advanced Topics**	Security challenges in cloud computing, Identity and Access Management (IAM), Data protection: encryption, integrity, backups, Privacy issues and compliance (GDPR, HIPAA), Multi-tenancy and isolation, Disaster recovery and fault tolerance, Edge computing and fog computing, Emerging trends: Kubernetes, OpenStack, Cloud Native applications	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Andrzej Goscinski, Cloud Computing: Principles and Paradigms,* Rajkumar Buyya, James Broberg, 1st Edition, 2013, Wiley
2. *Enterprise Cloud Computing: Technology, Architecture, Applications,* Gautam Shroff, 1st Edition, 2010, Cambridge University Press

Reference Books:

1. Anthony T. Velte, Toby J. Velte, *Cloud Computing: A Practical Approach,* 1st Edition, 2017, McGraw-Hill
2. Arshdeep Bahga & Vijay Madisetti, *Cloud Computing: A Hands-On Approach,* 2014, Universities Press

Paper IV/Subject Name: Wireless Sensor Networks		**Subject Code: CSE022D702**
Paper Type: PEC-IV		**Specialization: Network Engineering**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to introduce students to the architecture, components, and functioning of wireless sensor networks, localization, synchronization, and energy-efficient mechanisms, etc.

Prerequisites: Concepts of Networks and operating systems.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **understand** the architecture and components of wireless sensor networks.	**BT 1 & 2**
CO 2	**Apply** routing, MAC, and energy-efficient protocols in WSN applications.	**BT 3**
CO 3	**Analyze** the design challenges and communication protocols in WSNs.	**BT 4**
CO 4	**Evaluate** synchronization and localization algorithms for sensor nodes.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Introduction**	Overview of sensor networks: history, applications, and architecture, Components of WSN: sensor nodes, sink, gateway, Types of sensors and sensing models, Characteristics and challenges: scalability, fault tolerance, QoS, Comparison with ad hoc networks and traditional wireless networks, Node architecture: hardware components and operating systems (TinyOS, Contiki), Overview of IEEE 802.15.4 and ZigBee protocols	**22**

II	**Communication Protocols and Networking**	MAC protocols: S-MAC, T-MAC, B-MAC, IEEE 802.15.4, Energy-efficient MAC design, Routing in WSN: Data-centric routing: SPIN, Directed Diffusion, Hierarchical routing: LEACH, TEEN, APTEEN, Location-based routing: GEAR, GAF, Data aggregation and dissemination strategies, Cross-layer design issues	**22**
III	**Localisation, Synchronisation and Energy Management**	Localization techniques: range-based and range-free, Trilateration and multilateration, DV-Hop and centroid localization algorithms, Time synchronization in WSNs: TPSN, RBS, Power management and energy harvesting techniques, Sleep scheduling and adaptive duty cycling, Coverage and connectivity issues	**22**
IV	**Security and Applications**	Security issues: attacks on WSNs, trust models, and countermeasures, Encryption, authentication, and key management techniques, Quality of Service (QoS) in WSN, Applications: Environmental monitoring, Healthcare and wearable sensors, Industrial automation, Smart cities and smart agriculture, Military and surveillance systems, Case studies and research trends in IoT-enabled WSNs	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Wireless Sensor Networks: An Information Processing Approach,* Feng Zhao & Leonidas Guibas, 2004, Morgan Kaufmann
2. *Wireless Sensor Networks: Technology, Protocols, and Applications*, Kazem Sohraby, Daniel Minoli, & Taieb Znati, 1st Edition, 2007, Wiley

Reference Books:

1. Holger Karl & Andreas Willig, *Protocols and Architectures for Wireless Sensor Networks*, 1st Edition, 2007, Wiley
2. Bhaskar Krishnamachari, *Networking Wireless Sensors*, 2006, Cambridge University Press

Paper III/Subject Name: Embedded Systems and IOT		Subject Code: CSE022D703
Paper Type: PEC-III		Specialization: Semiconductors
L-T-P-C – 4-0-0-4	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to make the students understand the architecture and design of embedded systems used in IoT, familiarize students with microcontrollers, interfacing techniques, and programming., IoT protocols, cloud connectivity, and data analytics.

Prerequisites: Basics of Computer Architecture, Programming in C or Python, Basic Networking Concepts, Digital Logic Design

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the architecture and components of embedded systems and IoT.	**BT 1 & 2**
CO 2	**Apply** programming concepts to interface sensors and actuators with microcontrollers.	**BT 3**
CO 3	**Analyze** the functionality of communication protocols used in IoT applications.	**BT 4**
CO 4	**Evaluate** cloud integration, data analytics, and security mechanisms in IoT systems.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Introduction to embedded systems: definition, characteristics, applications, Embedded system architecture: hardware and software components, Types of embedded systems: standalone, real-time, networked, mobile, Introduction to IoT: definition, evolution, applications, Embedded vs IoT systems, IoT system architecture: sensing, network, data processing, application layers, Overview of IoT protocols stack (IoT-A, 6LoWPAN, CoAP, MQTT)	**22**
II	**Microcontrollers and Interfaces**	Microcontrollers: 8051, ARM Cortex basics, and comparison, Introduction to Arduino and Raspberry Pi, GPIO programming, timers, interrupts, Interfacing with sensors (temperature, humidity, motion, gas), Interfacing with actuators (motors, relays, buzzers), Serial communication: UART, I2C, SPI, Embedded C/Python for microcontroller programming	**22**
III	**Communication and Networking in IOT**	Communication models in IoT, Protocols: IEEE 802.15.4, Bluetooth, BLE, ZigBee, Z-Wave, LoRaWAN, Wi-Fi, MQTT, CoAP, HTTP/HTTPS, Network layer in IoT: IPv6, 6LoWPAN, RPL, Message queuing and broker models, Interoperability issues and IoT gateways, Case study: Smart Home Network Design	**22**

IV	**Cloud, Analytics and IOT Application Development**	Cloud platforms for IoT: AWS IoT, Google Cloud IoT, Azure IoT Hub, IoT analytics: data collection, storage, processing, Security and privacy in IoT: authentication, encryption, vulnerabilities, Power management and energy harvesting, Mobile and web interfaces for IoT control, Mini Project: design and implement a prototype (e.g., smart parking, home automation), Overview of IoT standards: OneM2M, IoTivity, AllJoyn	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Internet of Things: A Hands-On Approach,* Vijay Madisetti, Arshdeep Bahga, 1st Edition, 2015, Orient Black Swan
2. *Introduction to Embedded Systems*, Shibu K.V., 2nd Edition, 2017, McGraw-Hill

Reference Books:

1. Raj Kamal, *Internet of Things - Architecture and Design Principles*, 1st Edition, 2017, McGraw Hill
2. Jonathan W. Valvano, *Embedded Systems: Introduction to ARM Cortex-M Microcontrollers*, 2nd Edition, 2012, CreateSpace

Paper IV/Subject Name: VHDL		**Subject Code: CSE022D704**
Paper Type: PEC-IV		**Specialization: Semiconductors**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand digital system design using VHDL as a hardware description language, model combinational and sequential circuits using different VHDL modeling styles, simulate and synthesize digital designs for FPGAs and ASICs, etc.

Prerequisites: Digital Logic Design, Basics of Computer Architecture and Logic Programming

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **understand** the syntax, semantics, and data types of VHDL.	**BT 1& 2**
CO 2	**Model** and simulate combinational and sequential logic circuits using VHDL.	**BT 3**

CO 3	**Analyze** VHDL design styles: dataflow, behavioral, and structural.	**BT 4**
CO 4	**Evaluate** timing and synthesis constraints using simulation tools.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course Contents	Hours
I	**Introduction**	Overview of hardware description languages, VHDL design flow and applications, VHDL program structure: entity, architecture, library, Lexical elements: identifiers, literals, operators, Data types: bit, boolean, std_logic, arrays, records, enumerated types, Modeling styles: Behavioral modelling, Dataflow modelling, Structural modeling	**22**
II	**Programming for Combinational Logic**	Concurrent signal assignment and sequential statements, Conditional and selected signal assignment, Logic gates, adders, multiplexers, decoders, Comparators, encoders, ALU modules, Use of packages, functions, and procedures, Testbenches for combinational logic, Simulation using ModelSim or Vivado	**22**
III	**Sequential Logic Design**	Processes and sensitivity list, Variables vs signals, Flip-flops, registers, counters, Finite State Machines: Mealy and Moore models, Synchronous and asynchronous sequential circuits, Clocking and timing analysis, FSM design case studies using VHDL	**22**
IV	**Synthesis and Implementation**	Synthesis concepts and constraints, Introduction to FPGA and CPLD architectures, Mapping VHDL code to hardware, RTL vs Gate-level modelling, Timing analysis and optimization, Introduction to VHDL simulation tools (ModelSim, Vivado, Quartus), Mini Project: Implementing a VHDL-based digital design on FPGA	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *VHDL: Programming by Example,* Douglas L. Perry, 4th Edition, 2002, McGraw-Hill
2. *Fundamentals of Digital Logic with VHDL Design,* Stephen Brown & Zvonko Vranesic, Subsequent Edition, 1997, *McGraw-Hill*

Reference Books:

1. Bhasker J., *A VHDL Primer*, 3RD Edition, 1998, PHI Learning

Paper III/Subject Name: Introduction to Data Science		**Subject Code: CSE022D705**
Paper Type: PEC-III		**Specialization: General**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain the students about the mathematical concepts required for data science and teach data analytics problem solving frameworks.

Prerequisites: Basic concepts of Mathematics, Computer Programming, Data Structures and Databases

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Illustrate** a flow process for data science problems and **classify** data science problems into standard typology	**BT 2**
CO 2	**Construct** R or Python codes for data science solutions	**BT 3**
CO 3	**Examine** results to the solution approach	**BT 4**
CO 4	**Assess** the solution approach and Construct use cases to validate approach and identify modifications required	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course content**	**Periods**
I	**Introduction**	Introduction to Data Science , Data Analytics, Big Data ,Areas and Application of Data Sciences, Mathematical foundation of Data Science , descriptive statistics, notion of probability, distributions, mean, variance, covariance, covariance matrix, understanding univariate and multivariate normal distributions, introduction to hypothesis testing, confidence interval for estimates ,Statistical Inference:, Introduction to R and Python : Import –Export functions, DPLYR functions , Data Visualization .	**30**
II	**Algorithms for Data Sciences**	Introduction to machine learning, Linear regression and regularization, Model selection and evaluation, Classification: kNN, decision trees, Classification: SVM, Ensemble methods: random forests, Intro to probability: Naïve Bayes and logistic regression, Clustering: k-means, hierarchical clustering	**30**
III	**Recommendation Systems**	Algorithms for Recommendation Engine, Dimensionality Reduction, Singular Value Decomposition, Principal Component Analysis , Case Study	**14**
IV	**Information retrieval and Data Visualization**	Text mining and information retrieval :Mining Social-Network Graphs, Social networks as graphs, Clustering of graphs, Direct discovery of communities in graphs ,Partitioning of graphs, Neighbourhood properties in graphs ,Data Visualization, Basic principles, ideas and tools for data visualization	**14**
Total			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Doing Data Science, Straight Talk From The Frontline*, Cathy O'Neil and Rachel Schutt, 1st Edition, 2014, O'Reilly

Reference Books:

1. James, G., Witten, D., Hastie, T., Tibshirani, R., *An introduction to statistical learning with applications in R,* 7th Edition, 2014, Springer.
2. Murphy, K, *Machine Learning, A Probabilistic Perspective,* 2012, MIT Press.
3. Mohammed J. Zaki and Wagner Miera Jr., *Data Mining and Analysis: Fundamental Concepts and Algorithms*, 1st Edition, 2014, Cambridge University Press.

Paper IV/Subject Name: Fundamentals of Quantum Computing		**Subject Code: CSE022D706**
Paper Type: PEC-IV		**Specialization: General**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to provide the students with an overall insight to the core principles of quantum mechanics relevant to computing, quantum bits, gates, and circuits, major quantum algorithms and their applications and limitations, complexity, and error correction in quantum computation.

Prerequisites: Linear Algebra and Probability, Discrete Mathematics, Basics of Algorithms and Digital Logic.

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and i**llustrate** fundamental concepts of quantum mechanics as applied to computation.	**BT 1 & 2**
CO 2	**Experiment with** and simulate quantum bits and quantum gates.	**BT 3**
CO 3	**Analyze** quantum circuits and evaluate algorithmic performance.	**BT 4**
CO 4	**Assess** key quantum algorithms and their computational advantages.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**

I	**Introduction**	Classical vs Quantum computation, Postulates of quantum mechanics, Qubits and quantum states, Bra-ket notation, superposition, Quantum measurement and probability amplitudes, Tensor products and multi-qubit systems, Quantum entanglement	**22**
II	**Quantum Gates and Circuits**	Quantum logic gates: Pauli (X, Y, Z), Hadamard, Phase, T gate, Controlled gates: CNOT, Toffoli, Unitary operators and reversibility, Building quantum circuits from basic gates, Quantum circuit simulation using Qiskit or Quirk, Bell states and quantum teleportation protocol	**22**
III	**Quantum Algorithms**	Deutsch-Jozsa algorithm, Grover's search algorithm, Quantum Fourier Transform, Shor's factoring algorithm (overview only), Quantum parallelism and speed-up, Complexity comparison with classical algorithms, Quantum algorithm simulation using IBM Q Experience	**22**
IV	**Error Correction and Computing Platforms**	Sources of quantum noise and decoherence, Basic quantum error correction principles, The 3-qubit bit-flip and phase-flip codes, Quantum fault tolerance (overview), Introduction to IBM Q, Microsoft Q#, and Google Cirq, Quantum supremacy and future outlook, Applications: cryptography, machine learning, chemistry, optimization	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Quantum Computation and Quantum Information,* Michael A. Nielsen & Isaac L. Chuang, 2010, Cambridge University Press

Reference Books:

1. N.P. Stroud, *Quantum Mechanics for Electrical Engineers,* 1st Edition, 2012, CRC Press
2. Eleanor G. Rieffel & Wolfgang Polak, *Quantum Computing: A Gentle Introduction*, Illustrated Edition, 2014, MIT Press

6.9 Detailed Syllabus of 8th Semester

Paper I/Subject Name: Introduction to Machine Learning		Subject Code: CSE022C801
L-T-P-C – 3-0-2-4	Credit Units: 04	Scheme of Evaluation: TP

Objective:

The objectives of the course are to make the students learn fundamental concepts and types of machine learning, provide hands-on experience in building and evaluating ML models, develop the ability to choose appropriate algorithms for real-world tasks, etc.

Prerequisites: Linear Algebra, Probability and Statistics, Programming (Python preferred), Data Structures and Algorithms

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **Interpret** the basic concepts, types, and steps involved in machine learning.	**BT 1 & 2**
CO 2	**Apply** supervised and unsupervised learning algorithms to solve simple problems.	**BT 3**
CO 3	**Analyze** model performance using evaluation metrics	**BT 4**
CO 4	**Evaluate** various ML algorithms based on their suitability and accuracy.	**BT 5**

Detailed Syllabus

Modules	Topics	Course Contents	Hours
I	**Introduction**	What is Machine Learning?, Types of learning: Supervised, Unsupervised, Semi-supervised, Reinforcement Learning, ML workflow: data preprocessing, feature selection, model building, Overfitting and underfitting, Cross-validation and bias-variance tradeoff, Performance metrics: accuracy, precision, recall, F1-score, ROC, AUC	**18**
II	**Supervised Learning**	Linear Regression: cost function, gradient descent, Logistic Regression, Decision Trees and Random Forests, k-Nearest Neighbors (k-NN), Support Vector Machines (SVMs), Naïve Bayes classifier, Hands-on using scikit-learn (Python)	**18**
III	**Unsupervised Learning**	Clustering vs Classification, K-Means clustering, Hierarchical clustering, DBSCAN, Principal Component Analysis (PCA), Anomaly detection basics, Visualization techniques for clusters and reduced dimensions	**15**
IV	**Model Selection, Tuning, Application**	Model selection: train/test split, k-fold cross-validation, Hyperparameter tuning: Grid search and Random search, Ensemble methods: Bagging, Boosting (AdaBoost, Gradient Boosting), Introduction to deep learning (basic ANN structure	**15**

		only), ML applications: healthcare, finance, NLP, recommendation systems, Ethics and fairness in ML	
TOTAL			**66**

Introduction to Machine Learning Lab Syllabus

Detailed Syllabus:

Total Lab Hours for the semester = 30 (2 hours per week)

Minimum 10 Laboratory experiments based on the following-

1. Numpy, Pandas, Matplotlib, Seaborn, Scikit-learn overview, Importing datasets and basic exploration
2. Handling missing data, outliers, label encoding, Normalization, standardization, and feature scaling
3. Predicting continuous outcomes (e.g., housing prices), Visualizing regression line and residuals
4. Binary classification problem (e.g., tumor detection or pass/fail), Confusion matrix, precision, recall, F1-score
5. Building tree-based models, Visualizing decision boundaries and interpreting feature importance
6. Distance metrics and K selection, Classification and performance evaluation
7. Linear and kernel-based SVM, Visualizing hyperplanes and margin
8. Text classification (e.g., spam detection), Bag-of-words and TF-IDF vectorization
9. Unsupervised Learning – K-Means Clustering, Clustering unlabeled data
10. Elbow method and silhouette score,
11. Dimensionality Reduction – PCA, Reducing feature space
12. Visualizing 2D projections of high-dimensional data
13. Model Evaluation and Hyperparameter Tuning, Train/test split, k-fold cross-validation, GridSearchCV, RandomizedSearchCV
14. Mini Project: Apply full ML workflow (preprocessing, modeling, tuning, and evaluation) on a real-world dataset (e.g., Titanic, UCI dataset, Kaggle dataset

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
3* 22 NCH = 66 NCH	1 * 15 NCH = 30 NCH	8 * 3 NCH = 24 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Machine Learning,* Tom M. Mitchell, 1st Edition, 2017, McGraw-Hill
2. *Introduction to Machine Learning,* Ethem Alpaydin, 4th Edition, 2020, MIT Press

Reference Books:

1. Aurélien Géron, *Hands-On Machine Learning with Scikit-Learn*, 2nd Edition, 2019, Keras & TensorFlow, O'Reilly

Paper II/Subject Name: Cryptography and Network Security		Subject Code: CSE022C802
L-T-P-C – 4-0-0-3	Credit Units: 04	Scheme of Evaluation: T

Objective:

The objectives of the course are to explain the basics of cryptography, kinds of security threats in networks and to learn to find the vulnerabilities in programs and to overcome them and to teach about the models and standards for security.

Prerequisites: Concepts of Number Theory and Networking

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Understand** and **illustrate** basic cryptographic algorithms, message and web authentication and security issues.	**BT 2**
CO 2	**Demonstrate** the current legal and ethical issues towards information.	**BT 2**
CO 3	**Identify** the applications of different protocol like SSL, TLS etc.	**BT 3**
CO 4	**Analyze** and **assess** the security services and mechanisms	**BT 4**

Detailed Syllabus

Modules	Topics	Course Contents	Hours
I	**Introduction**	Need for Security, Security Approaches, Principles of Security, Types of Attacks, Brute Force Attack, Encryption, Decryption, Crptosystem, Cryptographic Techniques: Substitution Ciphers, Transposition Ciphers, Product Ciphers, Stegenography, Block Cipher, Stream Cipher.	**22**
II	**Symmetric and Asymmetric Key Cryptography**	Overview, Algorithm Modes and Types, Data Encryption Standard: Simplified DES, The Strength of DES, Differential and Linear Cryptanalysis. Triple DES, Blowfish. Confidentiality using Conventional Encryption: Placement of Encryption Function, Traffic Confidentiality, Key Distribution, Random Number Generation. Modular Arithmetic, Public Key Cryptography and RSA: Principles of Public Key Cryptosystems, Difference with Symmetric Key Cryptography, The RSA Algorithms, Key Management, Diffie Hellman Key Exchange.	**22**
III	**Authentication Protocols**	Message Authentication: Authentication Requirements, Authentication Functions, Message Authentication Codes, MD5 Message Digest Algorithms, Digital Signatures and Authentication Protocols: Digital Signatures, Authentication Protocols, Digital Signature Standards.	**22**
IV	**Security Protocols**	Security Applications and Protocols- Authentication Applications: Secure HTTP, HTTPS, ERT, SSH, Kerberos. Email Security: PGP, S/MIME. IP Security: Overview, IPSec architecture.	**22**

TOTAL	88

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Cryptography and Network Security*, Atul Kahate, 2nd Edition. 2003, Tata McGraw Hill.
2. *Cryptography and Network security,* Fourozan, 3rd Edition, 2007, McGraw Hill

Reference Books:

1. William Stallings, *Cryptography and Network Security*: *Principles and Practices*", 5th Edition, 2010, Prentice Hall.
2. Michael Howard, David LeBlanc, John Viega, *24 Deadly Sins of Software Security: Programming Flaws and How to Fix Them*, 1st Edition, 2009, Mc Graw Hill Osborne Media.

Paper III/Subject Name: Internet of Things		**Subject Code: CSE022D801**
Paper Type: PEC-V		**Specialization: Network Engineering**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand foundational knowledge of the Internet of Things and its architecture, IoT protocols and communication models, the integration of sensors, microcontrollers, and cloud platforms, real-world IoT applications and hands-on system design, etc.

Prerequisites: Basics of Computer Networks, Digital Electronics, Programming in Python or C

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Explain** and **understand** the basic concepts, architecture, and enabling technologies of the Internet of Things.	**BT 1 & 2**
CO 2	**Identify** the use of microcontrollers (e.g., Arduino/Raspberry Pi) in designing simple IoT applications.	**BT 3**
CO 3	**Analyze** data acquisition and processing techniques in IoT-based systems.	**BT 4**
CO 4	**Assess** the performance, security, and scalability of different IoT applications across domains.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction**	Definition and Evolution of IoT, IoT Ecosystem: Components and Architecture, Characteristics and Design Principles of IoT, Embedded Systems vs. IoT, IoT Levels: Perception, Network, Middleware, Application, Physical and Logical Design of IoT, IoT Enabling Technologies: Cloud Computing, Big Data, AI, Edge Computing	**22**
II	**IOT Hardware and Sensors**	Microcontrollers and Microprocessors (Arduino, Raspberry Pi, ESP32), Digital and Analog Sensors: Working and Interfacing, Actuators and their Types, GPIO, PWM, ADC, I2C, SPI basics, Sensor Data Acquisition and Conditioning, Power Sources and Management for IoT Devices, Interfacing Sensors with Microcontrollers (Hands-on with Arduino or Pi)	**22**
III	**Networking and Communication Protocols**	Network Layer & Communication Models, OSI Layer & Role in IoT, Protocols: HTTP, MQTT, CoAP, LoRa, Zigbee, BLE, NFC, IP Addressing in IoT (IPv6, 6LoWPAN), Cloud Integration for IoT Data (ThingSpeak, Blynk, Firebase, AWS IoT), Security Concerns: Encryption, Authentication, Privacy, Hands-on: Sending data to cloud via MQTT/HTTP	**22**
IV	**IOT Applications and Challenges**	Domain Applications: Smart Agriculture, Smart Home / Smart Cities, Healthcare IoT, Industrial IoT (IIoT) Case Studies: Google Nest, Amazon Echo, Smart Irrigation Challenges in IoT: Scalability, Interoperability, Security, Ethics Final Project Design and Development, Documentation and Presentation of IoT Prototype	**22**
Total			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Internet of Things: A Hands-On Approach,* Vijay Madisetti, Arshdeep Bahga, 1st Edition, 2015, Orient Black Swan

Reference Books:

1. Waltenegus Dargie, Christian Poellabauer, *Fundamentals of Wireless Sensor Networks: Theory and Practice*, 1st Edition, 2015, Wiley india Pvt. Ltd
2. Raj Kamal, *Internet of Things - Architecture and Design Principles*, 1st Edition, 2017, McGraw Hill

Paper III/Subject Name: Nanoelectronics		**Subject Code: CSE022D802**
Paper Type: PEC-		**Specialization: Semiconductors**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to make the students understand the principles and physics behind electronic behavior at the nanoscale, concept of quantum mechanics as applied to nano-devices, explore nano-scale materials, device fabrication, and measurement techniques, etc.

Prerequisites: Semiconductor Devices, Engineering Physics, Basic Knowledge of Electronic Devices and Circuits

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Describe** and **explain** the principles of quantum mechanics and their role in nano-scale electronic behavior.	**BT 1 & 2**
CO 2	**Apply** concepts of tunneling, quantization, and transport in nanoelectronic devices.	**BT 3**
CO 3	**Analyze** the electrical properties of nano-materials and quantum devices	**BT 4**
CO 4	**Evaluate** different fabrication and characterization techniques for nano-scale devices.	**BT 5**

Detailed Syllabus:

Modules	**Topics**	**Course Contents**	**Hours**
I	**Introduction**	Introduction to Nanoelectronics and Scaling Limits, Classical vs Quantum Regime, Overview of Nanomaterials: Nanotubes, Graphene, Nanowires, Quantum Dots, Energy bands and density of states in 1D, 2D, and 3D systems, Size effects and surface-to-volume ratio, Ballistic vs diffusive transport, Overview of quantum confinement	**22**
II	**Quantum Transport and Tunneling Phenomena**	Introduction to quantum mechanics for nanodevices, Tunneling current and potential barriers, Quantum tunneling: Tunneling diodes, quantum wells, Landauer formalism for quantum transport, Coulomb blockade and single electron transistor (SET), Quantum capacitance and inductance, Nano-MOSFET characteristics	**22**
III	**Fabrication and Characterization Techniques**	Lithography: Photolithography, Electron-beam lithography, Nanoimprint, Deposition techniques: CVD, PVD, ALD, Etching: Wet and dry etching methods, Scanning Electron Microscopy (SEM), AFM, STM, Electron Transport Measurement Techniques, Cleanroom practices and nanofabrication safety	**22**

IV	**Emerging Devices and Applications**	CNT-FETs, FinFETs, and Tunnel FETs, Spintronics and Spin-based devices, Molecular electronics, Quantum dots in computing and bio-sensing, Nano-memristors and neuromorphic computing, Nanoelectronic memory and logic devices, Applications in energy, healthcare, and IoT	**22**
TOTAL			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Books:

1. *Introduction to Nanoelectronics: Science, Nanotechnology, Engineering, and Applications,* V. Mitin, V. Kochelap, M. Stroscio, 2008, Cambridge University Press
2. *Nanoelectronics: Principles & Devices,* K.K. Chauhan, 2nd Edition, 2008, Oxford University Press

Reference Books:

1. George W. Hanson, Fundamentals of Nanoelectronics, 1st Edition, 2009, Pearson

Paper III/Subject Name: Introduction to Cybersecurity		**Subject Code: CSE022D803**
Paper Type: PEC-V		**Specialization: General**
L-T-P-C – 4-0-0-4	**Credit Units: 04**	**Scheme of Evaluation: T**

Objective:

The objectives of the course are to explain the students about the fundamentals and importance of cybersecurity, security threats, vulnerabilities, and cryptographic principles, security protocols, access control, and network defense mechanisms, develop awareness of legal, ethical, and emerging cybersecurity issues.

Prerequisites: Basics of Networking, Operating Systems, Basic understanding of Data Structures and Internet usage

Course Outcomes

On successful completion of the course the students will be able to:		
SI No	**Course Outcome**	**Blooms Taxonomy Level**
CO 1	**Illustrate** the basic concepts of cybersecurity, threats, and security principles.	**BT 2**
CO 2	**Apply** cryptographic and access control techniques to protect data.	**BT 3**
CO 3	**Examine** various types of security vulnerabilities and attacks.	**BT 4**
CO 4	**Assess** security mechanisms in operating systems, networks, and web applications.	**BT 5**

Detailed Syllabus:

Modules	Topics	Course content	Periods
I	**Introduction**	Definition and need for cybersecurity, Cybercrime and cyberwarfare, Threats: malware, phishing, ransomware, DoS/DDoS, spoofing, Vulnerabilities and risk analysis, CIA triad (Confidentiality, Integrity, Availability), Security goals, principles of security, and types of attackers, Basics of ethical hacking and penetration testing	**22**
II	**Cryptography and Authentication**	Basic principles of cryptography, Symmetric vs Asymmetric encryption, DES, AES, RSA: working and applications, Hash functions: MD5, SHA family, Digital signatures and certificates, Authentication techniques: passwords, OTP, biometrics, Public Key Infrastructure (PKI)	**22**
III	**Network and Security Systems**	Security in OS and file systems, Secure software development, Network security protocols: SSL/TLS, HTTPS, IPsec, Firewalls and Intrusion Detection/Prevention Systems (IDS/IPS), Web application security: SQL injection, XSS, CSRF, Secure communication: VPNs, email security (PGP), Case studies: recent attacks and mitigation	**22**
IV	**Cyber laws, Ethics and Emerging Trends**	Introduction to Cyber Laws (India IT Act, GDPR overview), Intellectual property rights, privacy, and data protection, Cyber ethics and digital forensics basics, Security audits and compliance (ISO 27001, SOC 2), Security in IoT, cloud, mobile, AI systems, Emerging areas: zero trust security, blockchain in cybersecurity, Cybersecurity careers and certifications	**22**
Total			**88**

Credit Distribution		
Lecture/ Tutorial	**Practicum**	**Experiential Learning**
4* 22 NCH = 88 NCH	--	8 * 4 NCH = 32 NCH (Problem Solving, Seminar, Case Study, Discussion, Internship, Projects)

Text Book:

1. *Computer Security: Principles and Practice,* William Stallings, 4th Edition, 2017, Pearson

Reference Books:

1. Behrouz A. Forouzan, *Cryptography and Network Security*, 2007, McGraw-Hill
2. Michael T. Goodrich, *Introduction to Computer Security*, Pearson